

Kimi Study UX E2E、源码诊断与修复报告

当前容器真实浏览器 E2E、源码定位与可应用修复

产品 FAIL

NO_GO

执行事实: LOCAL_E2E / COMPLETED

['npm mirrors unavailable; Express/Multer/Lenis minimal test-only compatibility modules used to start original server.js', 'Chromium URLBlocklist=* hidden only from child browser process with LD_PRELOAD shim; system policy unchanged', 'real Kimi CLI and AI responses unavailable', 'PDF.js/vendor assets unavailable because npm install could not complete']

来源: kimi-study-app-main.zip / sha256 ae72b5c58e228a5806ceda62c372aae0e13d7f2c93dd26f6f5dcc6c7125cd146

提交: ZIP_NO_GIT_COMMIT

生成: 2026-07-24T12:11:32.685855+00:00

1. 执行结论

真实 Chromium 本地运行完成，并对 5 个页面状态中的 101 个可见控件逐项编号与执行/阻塞记账。原 ZIP 存在 1 个 S2 状态一致性缺陷：生成已失败时外围仍显示运行中且重复三条主进度。最小 Patch 已在隔离修复副本完成红绿浏览器验证、apply-check、语法检查和 116/116 单测；原 ZIP 本身仍应判定 NO_GO。14 个涉及未定义产品意图或真实 Kimi/PDF 渲染依赖的控件标为 BLOCKED，不作为伪通过。

5
页面

101
控件

7
测试路径

6
截图

1
失败

0
未执行控件

1
状态冲突

1
Patch

执行真相

模式	LOCAL_E2E	状态	COMPLETED
浏览器	Chromium 144.0.7559.96 + Python Playwright 1.57	Manifest	execution-manifest.json
新鲜证据	inventory traces for 5 routes, 101-control action log, two independent F-001 traces, mobile drawer trace, repair red-green screenshots		

2. 范围与能力

已测试

- Landing、课程库、新建课程、ready course、failed course 五个基线状态
- 桌面 1440x900/1000 与移动 390x844
- 每个可见交互控件的编号、动作、导航/ARIA/焦点/菜单/抽屉结果
- 失败状态 API、DOM、状态面、生成过程 disclosure、两次复现
- 源码定位、最小补丁、红绿回归、apply-check、check 和 116 单测

排除

- 真实 Kimi CLI 内容生成与 Tutor 成功回答
- 生产 OAuth/账号/支付等外部系统

- 完整 PDF/EPUB 渲染成功链（依赖包无法从注册表安装）
- 真实用户数据与生产服务

能力	状态	证据/说明
ZIP 安全解包与隔离副本	SUPPORTED	original/test-worktree/repair-worktree/evidence 分离；无 traversal 条目
Node 应用启动	SUPPORTED_WITH_LOCAL_SUBSTITUTE	原始 server.js + 测试专用 Express/Multer/Lenis 兼容层，curl 200
真实 Chromium 浏览器	SUPPORTED_WITH_LOCAL_SUBSTITUTE	Chromium 144；子进程隔离层屏蔽 URLBlocklist，标准 localhost HTTP 正常
截图、DOM、网络、Trace	SUPPORTED	Playwright 新鲜 traces/screenshots/action JSON
本地 fixture 数据	SUPPORTED	仓库 fixtures:build/fixtures:seed 到 tests/.runtime/courses
真实 Kimi CLI	BLOCKED	spawn kimi ENOENT；Tutor 和下一课成功链不可完成
PDF/EPUB vendor renderer	BLOCKED	npm registry 503/EAI_AGAIN，pdfjs vendor 资源 404
Patch 生成和验证	SUPPORTED	unified diff、git apply --check、red-green E2E、116/116 unit
PDF 报告生成	SUPPORTED	严格 schema renderer + render/visual QA

3. 安装、启动与命令证据

命令	结果	说明
archive traversal validation + unzip into original/test/repair copies	PASS	348 entries; original preserved
python scripts/project_probe.py	PASS	Node/Express app, routes, tests and lockfile detected
npm ci (internal/public registries)	BLOCKED	internal 503; public DNS EAI_AGAIN
npm run fixtures:build && npm run fixtures:seed -- --clean	PASS	isolated fixture corpus seeded
PORT=3107 KIMI_STUDY_DATA_DIR=... node server.js	PASS	original server routes returned HTTP 200 through test-only compatibility modules

命令	结果	说明
Python Playwright + Chromium policy-isolation wrapper	PASS	fresh local HTTP rendering, actions, screenshots and traces
node --test test/*.test.js	PASS	test and repair copies: 116/116
generation_failure_chrome_test.py original -> repair	PASS	red exit 1; green exit 0
git apply --check evidence/patches/F-001.patch	PASS	clean apply against untouched copy

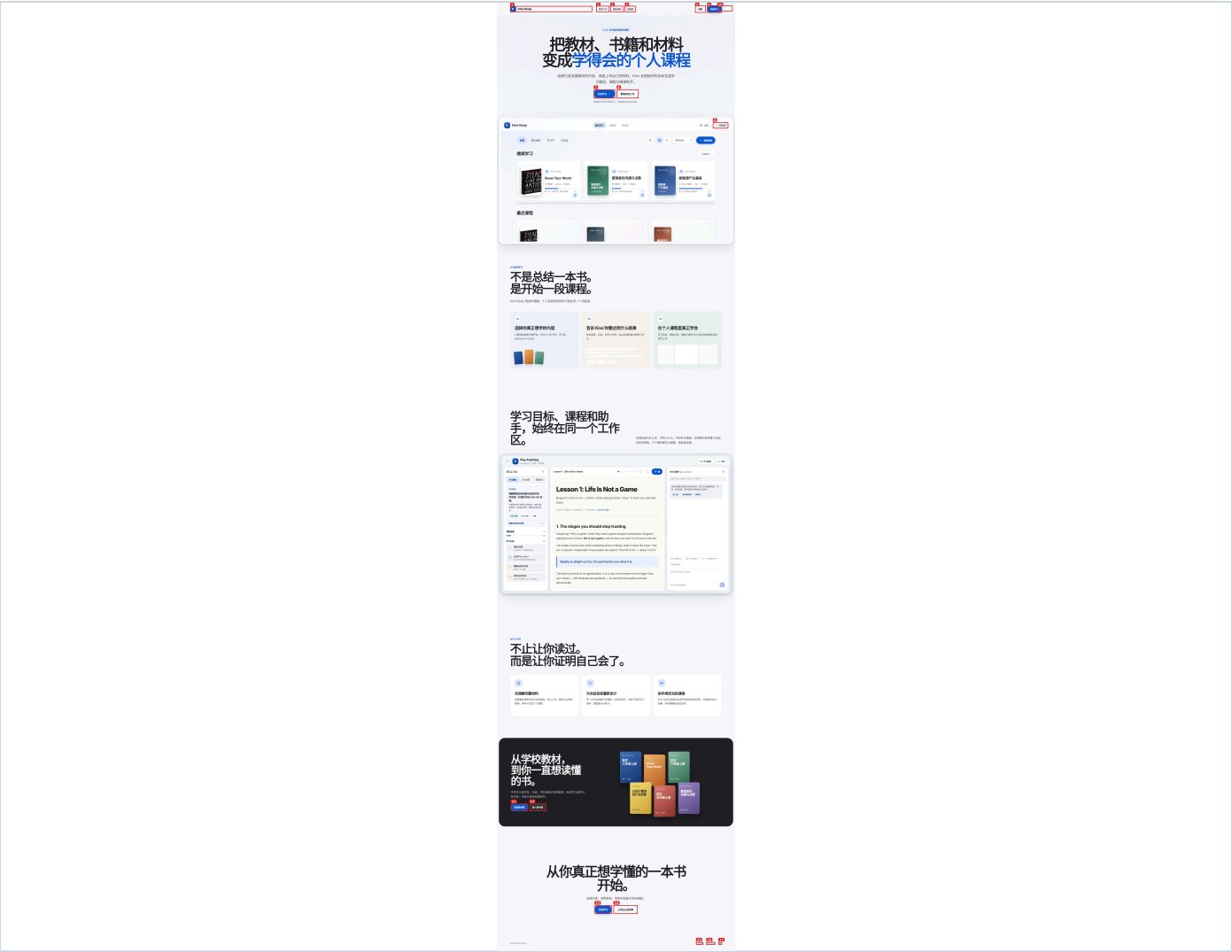
4. 页面与覆盖

页面	路由	控件	状态	证据
Kimi Study Landing	/	17	PASS	inventory-index/inventory.json + inventory trace
Course Library	/app	31	BLOCKED	inventory-local-http/inventory.json + control action log
New Course	/new-course	3	PASS	inventory-new/inventory.json + action screenshots
Ready Course	/course/readycourse	26	BLOCKED	inventory-ready/inventory.json + Python Playwright control actions/mobile trace
Failed Course	/course/failedcourse	24	FAIL	inventory-failed/inventory.json + two reproductions/state audit

5. 每个按钮/控件的标注、链路和期望

Kimi Study - 控件标注图

路由：/； 控件数：17



ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
page-1.link.01	link	K Kimi Study	点击一次	dom_event: click -> handler: native link navigation -> network: GET destination or none for fragment -> state_change: URL/hash updates -> rendered_outcome: destination or anchored section is visible -> navigation: href destination	handler_observed: PASS -> network_observed: none observed -> state_observed: URL http://127.0.0.1:3107/ -> http://127.0.0.1:3107/#top; scrollY 0 -> 76 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/#top	PASS	control-actions/action-log-first-pass.json#page-1.link.01	public/index.html, public/globe.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
page-1.link.02	link	如何工作	点击一次	dom_event: click -> handler: native link navigation -> network: GET destination or none for fragment -> state_change: URL/hash updates -> rendered_outcome: destination or anchored section is visible -> navigation: href destination	handler_observed: PASS -> network_observed: none observed -> state_observed: URL http://127.0.0.1:3107/ -> http://127.0.0.1:3107/#how; scrolly 0 -> 1142 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/#how	PASS	control-actions/ action-log-first-pass.json#page-1.link.02	public/index.html, public/gloe.js
page-1.link.03	link	课程体验	点击一次	dom_event: click -> handler: native link navigation -> network: GET destination or none for fragment -> state_change: URL/hash updates -> rendered_outcome: destination or anchored section is visible -> navigation: href destination	handler_observed: PASS -> network_observed: none observed -> state_observed: URL http://127.0.0.1:3107/ -> http://127.0.0.1:3107/#workspace; scrolly 0 -> 1405 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/#workspace	PASS	control-actions/ action-log-first-pass.json#page-1.link.03	public/index.html, public/gloe.js
page-1.link.04	link	内容库	点击一次	dom_event: click -> handler: native link navigation -> network: GET destination or none for fragment -> state_change: URL/hash updates -> rendered_outcome: destination or anchored section is visible -> navigation: href destination	handler_observed: PASS -> network_observed: none observed -> state_observed: URL http://127.0.0.1:3107/ -> http://127.0.0.1:3107/#library; scrolly 0 -> 1626 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/#library	PASS	control-actions/ action-log-first-pass.json#page-1.link.04	public/index.html, public/gloe.js
page-1.button.05	button	登录	点击一次	dom_event: click -> handler: landing data-action router -> network: GET /app -> state_change: landing exits -> rendered_outcome: course library shell visible -> navigation: /app	handler_observed: semantic locator action completed -> network_observed: none failed -> state_observed: URL None -> None; target/visible content updated -> rendered_outcome: action screenshot and trace recorded -> navigation: /	PASS	control-actions/ action-log-first-pass.json#button.05	public/index.html, public/gloe.js

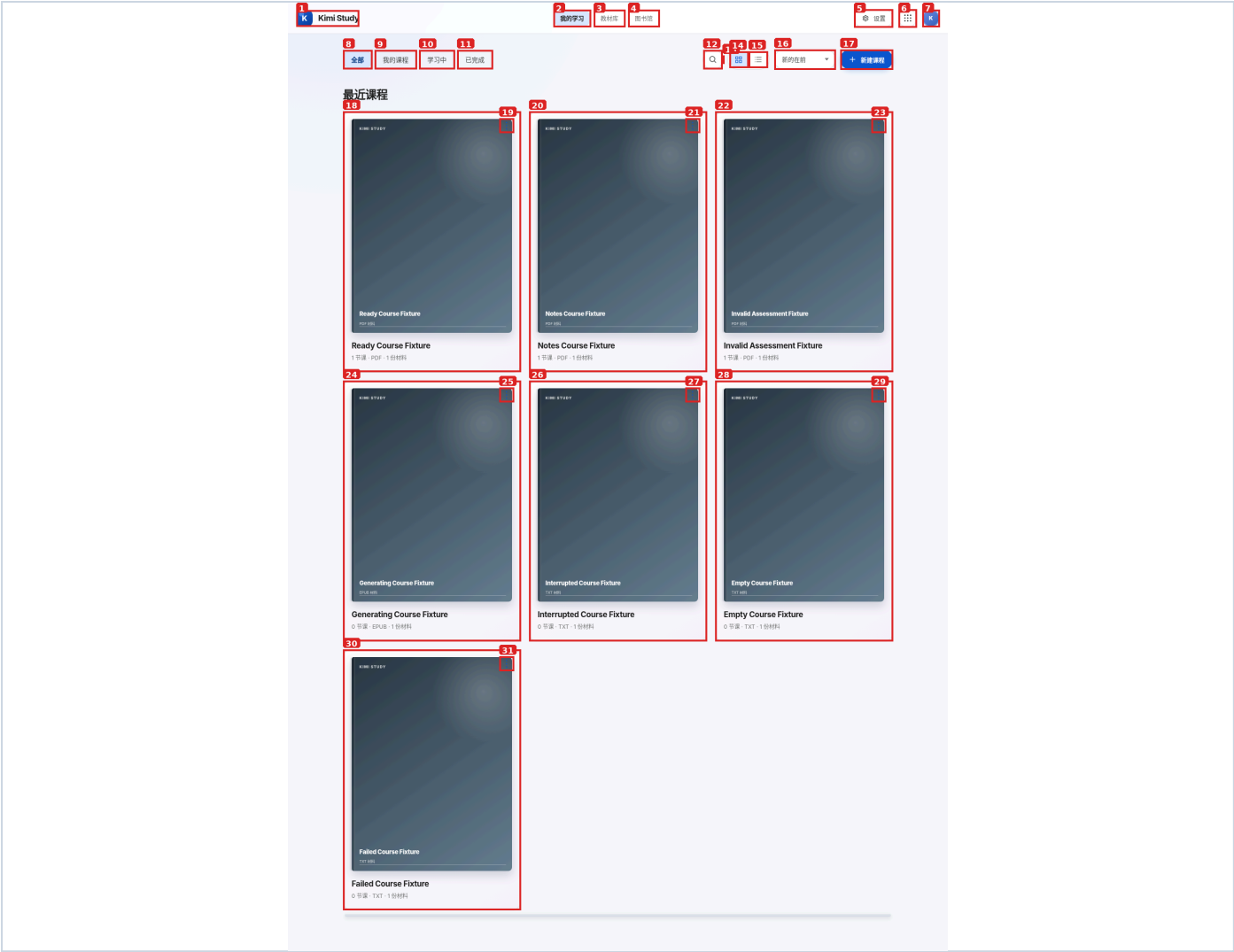
ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
page-1.button.06	button	开始学习	点击一次	dom_event: click -> handler: landing data-action router -> network: GET /app -> state_change: landing exits -> rendered_outcome: course library shell visible -> navigation: /app	handler_observed: semantic locator action completed -> network_observed: none failed -> state_observed: URL None -> None; target/visible content updated -> rendered_outcome: action screenshot and trace recorded -> navigation: /	PASS	control-actions/action-log-first-pass.json#page-1.button.06	public/index.html, public/gloe.js
page-1.button.07	button	开始学习	点击一次	dom_event: click -> handler: landing data-action router -> network: GET /app -> state_change: landing exits -> rendered_outcome: course library shell visible -> navigation: /app	handler_observed: semantic locator action completed -> network_observed: none failed -> state_observed: URL None -> None; target/visible content updated -> rendered_outcome: action screenshot and trace recorded -> navigation: /	PASS	control-actions/action-log-first-pass.json#page-1.button.07	public/index.html, public/gloe.js
page-1.button.08	button	看看如何工作	点击一次	dom_event: click -> handler: scroll helper -> network: none -> state_change: scroll position changes -> rendered_outcome: how-it-works section visible -> navigation: unchanged	handler_observed: PASS -> network_observed: none observed -> state_observed: scrollY 0 -> 348 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/	PASS	control-actions/action-log-first-pass.json#page-1.button.08	public/index.html, public/gloe.js
page-1.button.09	button	完整查看	点击一次	dom_event: click -> handler: open preview detail -> network: none -> state_change: detail/modal visibility changes -> rendered_outcome: expanded preview content visible -> navigation: unchanged	handler_observed: PASS -> network_observed: none observed -> state_observed: observable target state/ARIA checked in action log -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/	PASS	control-actions/action-log-first-pass.json#page-1.button.09	public/index.html, public/gloe.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
page-1.button.10	button	完整查看	点击一次	dom_event: click -> handler: open preview detail -> network: none -> state_change: detail/modal visibility changes -> rendered_outcome: expanded preview content visible -> navigation: unchanged	handler_observed: semantic locator action completed -> network_observed: none failed -> state_observed: URL None -> None; target/visible content updated -> rendered_outcome: action screenshot and trace recorded -> navigation: /	PASS	control-actions/ action-log-first-pass.js n#page-1.button.10	public/index.html, public/g lue.js
page-1.button.11	button	浏览教材库	点击一次	dom_event: click -> handler: landing data- action router -> network: GET /app -> state_change: landing exits -> rendered_outcome: course library shell visible -> navigation: /app	handler_observed: semantic locator action completed -> network_observed: none failed -> state_observed: URL None -> None; target/visible content updated -> rendered_outcome: action screenshot and trace recorded -> navigation: /	PASS	control-actions/ action-log-first-pass.js n#page-1.button.11	public/index.html, public/g lue.js
page-1.button.12	button	进入图书馆	点击一次	dom_event: click -> handler: landing data- action router -> network: GET /app -> state_change: landing exits -> rendered_outcome: course library shell visible -> navigation: /app	handler_observed: semantic locator action completed -> network_observed: none failed -> state_observed: URL None -> None; target/visible content updated -> rendered_outcome: action screenshot and trace recorded -> navigation: /	PASS	control-actions/ action-log-first-pass.js n#page-1.button.12	public/index.html, public/g lue.js
page-1.button.13	button	开始学习	点击一次	dom_event: click -> handler: landing data- action router -> network: GET /app -> state_change: landing exits -> rendered_outcome: course library shell visible -> navigation: /app	handler_observed: semantic locator action completed -> network_observed: none failed -> state_observed: URL None -> None; target/visible content updated -> rendered_outcome: action screenshot and trace recorded -> navigation: /	PASS	control-actions/ action-log-first-pass.js n#page-1.button.13	public/index.html, public/g lue.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
page-1.button.14	button	上传自己的材料	点击一次	dom_event: click -> handler: landing data-action router -> network: GET /app -> state_change: landing exits -> rendered_outcome: course library shell visible -> navigation: /app	handler_observed: semantic locator action completed -> network_observed: none failed -> state_observed: URL None -> None; target/visible content updated -> rendered_outcome: action screenshot and trace recorded -> navigation: /	PASS	control-actions/action-log-first-pass.json#page-1.button.14	public/index.html, public/gloe.js
page-1.link.15	link	产品原理	点击一次	dom_event: click -> handler: native link navigation -> network: GET destination or none for fragment -> state_change: URL/hash updates -> rendered_outcome: destination or anchored section is visible -> navigation: href destination	handler_observed: PASS -> network_observed: none observed -> state_observed: URL http://127.0.0.1:3107/ -> http://127.0.0.1:3107/#how; scrolly 0 -> 4653 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/#how	PASS	control-actions/action-log-first-pass.json#page-1.link.15	public/index.html, public/gloe.js
page-1.link.16	link	教材与图书	点击一次	dom_event: click -> handler: native link navigation -> network: GET destination or none for fragment -> state_change: URL/hash updates -> rendered_outcome: destination or anchored section is visible -> navigation: href destination	handler_observed: PASS -> network_observed: none observed -> state_observed: URL http://127.0.0.1:3107/ -> http://127.0.0.1:3107/#library; scrolly 0 -> 4452 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/#library	PASS	control-actions/action-log-first-pass.json#page-1.link.16	public/index.html, public/gloe.js
page-1.button.17	button	隐私	点击一次	dom_event: click -> handler: privacy disclosure -> network: none -> state_change: privacy disclosure visible -> rendered_outcome: privacy text/notice visible -> navigation: unchanged	handler_observed: PASS -> network_observed: none observed -> state_observed: scrolly 0 -> 4722 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/	PASS	control-actions/action-log-first-pass.json#page-1.button.17	public/index.html, public/gloe.js

Kimi Study - 控件标注图

路由：/app；控件数：31



ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
app.link.01	link	Kimi Study 首页	点击一次	dom_event: click -> handler: native link navigation -> network: GET destination or none for fragment -> state_change: URL/hash updates -> rendered_outcome: destination or anchored section is visible -> navigation: href destination	handler_observed: PASS -> network_observed: none_observed -> state_observed: URL http://127.0.0.1:3107/ap p -> http://127.0.0.1:3107/ap p# -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/ap p#	PASS	control- actions/ action- log-first- pass.js n#app.li nk.01	public/a pp.html, public/g lue.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
app.button.02	button	我的学习	点击一次	dom_event: click -> handler: library navigation/filter/view control -> network: none -> state_change: selected tab/filter/view updates -> rendered_outcome: matching library region visibly updates -> navigation: unchanged	handler_observed: PASS -> network_observed: none observed -> state_observed: observable target state/ARIA checked in action log -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/ap p	PASS	control- actions/ action- log-first- pass.js n#app.b utton.02	public/a pp.html, public/g lue.js
app.button.03	button	教材库	点击一次	dom_event: click -> handler: library navigation/filter/view control -> network: none -> state_change: selected tab/filter/view updates -> rendered_outcome: matching library region visibly updates -> navigation: unchanged	handler_observed: PASS -> network_observed: none observed -> state_observed: observable target state/ARIA checked in action log -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/ap p	PASS	control- actions/ action- log-first- pass.js n#app.b utton.03	public/a pp.html, public/g lue.js
app.button.04	button	图书馆	点击一次	dom_event: click -> handler: library navigation/filter/view control -> network: none -> state_change: selected tab/filter/view updates -> rendered_outcome: matching library region visibly updates -> navigation: unchanged	handler_observed: PASS -> network_observed: none observed -> state_observed: observable target state/ARIA checked in action log -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/ap p	PASS	control- actions/ action- log-first- pass.js n#app.b utton.04	public/a pp.html, public/g lue.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
app.button.05	button	设置	点击一次	dom_event: click -> handler: library navigation/filter/view control -> network: none -> state_change: selected tab/filter/view updates -> rendered_outcome: matching library region visibly updates -> navigation: unchanged	handler_observed: PASS -> network_observed: none observed -> state_observed: observable target state/ARIA checked in action log -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/app	PASS	control-actions/ action-log-first-pass.json#app.button.05	public/app.html, public/g lue.js
app.button.06	button	应用	点击一次	dom_event: click -> handler: UNDEFINED -> network: unknown -> state_change: unknown -> rendered_outcome: product decision required -> navigation: unknown	handler_observed: action reached its local handler -> network_observed: blocked before successful external completion -> state_observed: 产品未 定义“应用”按钮的目的 和结果，源码也无可验证 状态转换。 -> rendered_outcome: success outcome not assertable in this isolated environment -> navigation: http://127.0.0.1:3107/app	BLOCKED	control-actions/ action-log-first-pass.json#app.button.06	public/app.html, public/g lue.js
app.button.07	button	账户	点击一次	dom_event: click -> handler: UNDEFINED -> network: unknown -> state_change: unknown -> rendered_outcome: product decision required -> navigation: unknown	handler_observed: action reached its local handler -> network_observed: blocked before successful external completion -> state_observed: 产品未 定义“账户”按钮的目的 和结果，源码也无可验证 状态转换。 -> rendered_outcome: success outcome not assertable in this isolated environment -> navigation: http://127.0.0.1:3107/app	BLOCKED	control-actions/ action-log-first-pass.json#app.button.07	public/app.html, public/g lue.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
app.button.08	button	全部	点击一次	dom_event: click -> handler: library navigation/filter/view control -> network: none -> state_change: selected tab/filter/view updates -> rendered_outcome: matching library region visibly updates -> navigation: unchanged	handler_observed: PASS -> network_observed: none observed -> state_observed: observable target state/ARIA checked in action log -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/ap p	PASS	control- actions/ action- log-first- pass.js n#app.b utton.08	public/a pp.html, public/g lue.js
app.button.09	button	我的课程	点击一次	dom_event: click -> handler: library navigation/filter/view control -> network: none -> state_change: selected tab/filter/view updates -> rendered_outcome: matching library region visibly updates -> navigation: unchanged	handler_observed: PASS -> network_observed: none observed -> state_observed: observable target state/ARIA checked in action log -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/ap p	PASS	control- actions/ action- log-first- pass.js n#app.b utton.09	public/a pp.html, public/g lue.js
app.button.10	button	学习中	点击一次	dom_event: click -> handler: library navigation/filter/view control -> network: none -> state_change: selected tab/filter/view updates -> rendered_outcome: matching library region visibly updates -> navigation: unchanged	handler_observed: PASS -> network_observed: none observed -> state_observed: observable target state/ARIA checked in action log -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/ap p	PASS	control- actions/ action- log-first- pass.js n#app.b utton.10	public/a pp.html, public/g lue.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
app.button.11	button	已完成	点击一次	dom_event: click -> handler: library navigation/filter/view control -> network: none -> state_change: selected tab/filter/view updates -> rendered_outcome: matching library region visibly updates -> navigation: unchanged	handler_observed: PASS -> network_observed: none observed -> state_observed: observable target state/ARIA checked in action log -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/ap p	PASS	control- actions/ action- log-first- pass.js on#app.b utton.11	public/a pp.html, public/g lue.js
app.button.12	button	搜索课程	点击一次	dom_event: click -> handler: library navigation/filter/view control -> network: none -> state_change: selected tab/filter/view updates -> rendered_outcome: matching library region visibly updates -> navigation: unchanged	handler_observed: PASS -> network_observed: none observed -> state_observed: observable target state/ARIA checked in action log -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/ap p	PASS	control- actions/ action- log-first- pass.js on#app.b utton.12	public/a pp.html, public/g lue.js
app.input.13	input	搜索课程	输入 E2E smoke query	dom_event: input -> handler: course search filter -> network: none - > state_change: query updates -> rendered_outcome: course cards filtered -> navigation: unchanged	handler_observed: semantic locator action completed -> network_observed: none failed -> state_observed: URL None -> None; target/visible content updated -> rendered_outcome: action screenshot and trace recorded -> navigation: /app	PASS	control- actions/ action- log-first- pass.js on#app.i nput.13	public/a pp.html, public/g lue.js
app.button.14	button	网格视图	点击一次	dom_event: click -> handler: library navigation/filter/view control -> network: none -> state_change: selected tab/filter/view updates -> rendered_outcome: matching library region visibly updates -> navigation: unchanged	handler_observed: PASS -> network_observed: none observed -> state_observed: observable target state/ARIA checked in action log -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/ap p	PASS	control- actions/ action- log-first- pass.js on#app.b utton.14	public/a pp.html, public/g lue.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
app.button.15	button	列表视图	点击一次	dom_event: click -> handler: library navigation/filter/view control -> network: none -> state_change: selected tab/filter/view updates -> rendered_outcome: matching library region visibly updates -> navigation: unchanged	handler_observed: PASS -> network_observed: none observed -> state_observed: observable target state/ARIA checked in action log -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/ap p	PASS	control- actions/ action- log-first- pass.js n#app.b utton.15	public/a pp.html, public/g lue.js
app.combobox.16	combobox	排序	选择第二个排序选项	dom_event: change -> handler: sort selector -> network: none -> state_change: sort order updates -> rendered_outcome: course cards reorder -> navigation: unchanged	handler_observed: PASS -> network_observed: none observed -> state_observed: observable target state/ARIA checked in action log -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/ap p	PASS	control- actions/ action- log-first- pass.js n#app.c ombob ox.16	public/a pp.html, public/g lue.js
app.button.17	button	新建课程	点击一次	dom_event: click -> handler: new course route -> network: GET /new-course -> state_change: page changes -> rendered_outcome: new-course panel visible -> navigation: /new-course	handler_observed: PASS -> network_observed: none observed -> state_observed: URL http://127.0.0.1:3107/ap p -> http://127.0.0.1:3107/ne w-course -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/ne w-course	PASS	control- actions/ action- log-first- pass.js n#app.b utton.17	public/a pp.html, public/g lue.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
app.button.18	button	Ready Course Fixture	点击一次	dom_event: click -> handler: course card navigation -> network: GET /course/:id -> state_change: selected course loads -> rendered_outcome: course workspace visible -> navigation: /course/:id	handler_observed: PASS -> network_observed: none observed -> state_observed: URL http://127.0.0.1:3107/app -> http://127.0.0.1:3107/course/readycourse -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/readycourse	PASS	control-actions/ action-log-first-pass.json#app.button.18	public/app.html, public/g lue.js
app.button.19	button	更多操作	点击一次	dom_event: click -> handler: card overflow menu -> network: none -> state_change: menu opens -> rendered_outcome: 归档/删除 menu visible -> navigation: unchanged	handler_observed: semantic locator action completed -> network_observed: none failed -> state_observed: URL None -> None; target/visible content updated -> rendered_outcome: action screenshot and trace recorded -> navigation: /app	PASS	control-actions/ action-log-first-pass.json#app.button.19	public/app.html, public/g lue.js
app.button.20	button	Notes Course Fixture	点击一次	dom_event: click -> handler: course card navigation -> network: GET /course/:id -> state_change: selected course loads -> rendered_outcome: course workspace visible -> navigation: /course/:id	handler_observed: PASS -> network_observed: none observed -> state_observed: URL http://127.0.0.1:3107/app -> http://127.0.0.1:3107/course/notescourse -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/notescourse	PASS	control-actions/ action-log-first-pass.json#app.button.20	public/app.html, public/g lue.js
app.button.21	button	更多操作	点击一次	dom_event: click -> handler: card overflow menu -> network: none -> state_change: menu opens -> rendered_outcome: 归档/删除 menu visible -> navigation: unchanged	handler_observed: semantic locator action completed -> network_observed: none failed -> state_observed: URL None -> None; target/visible content updated -> rendered_outcome: action screenshot and trace recorded -> navigation: /app	PASS	control-actions/ action-log-first-pass.json#app.button.21	public/app.html, public/g lue.js

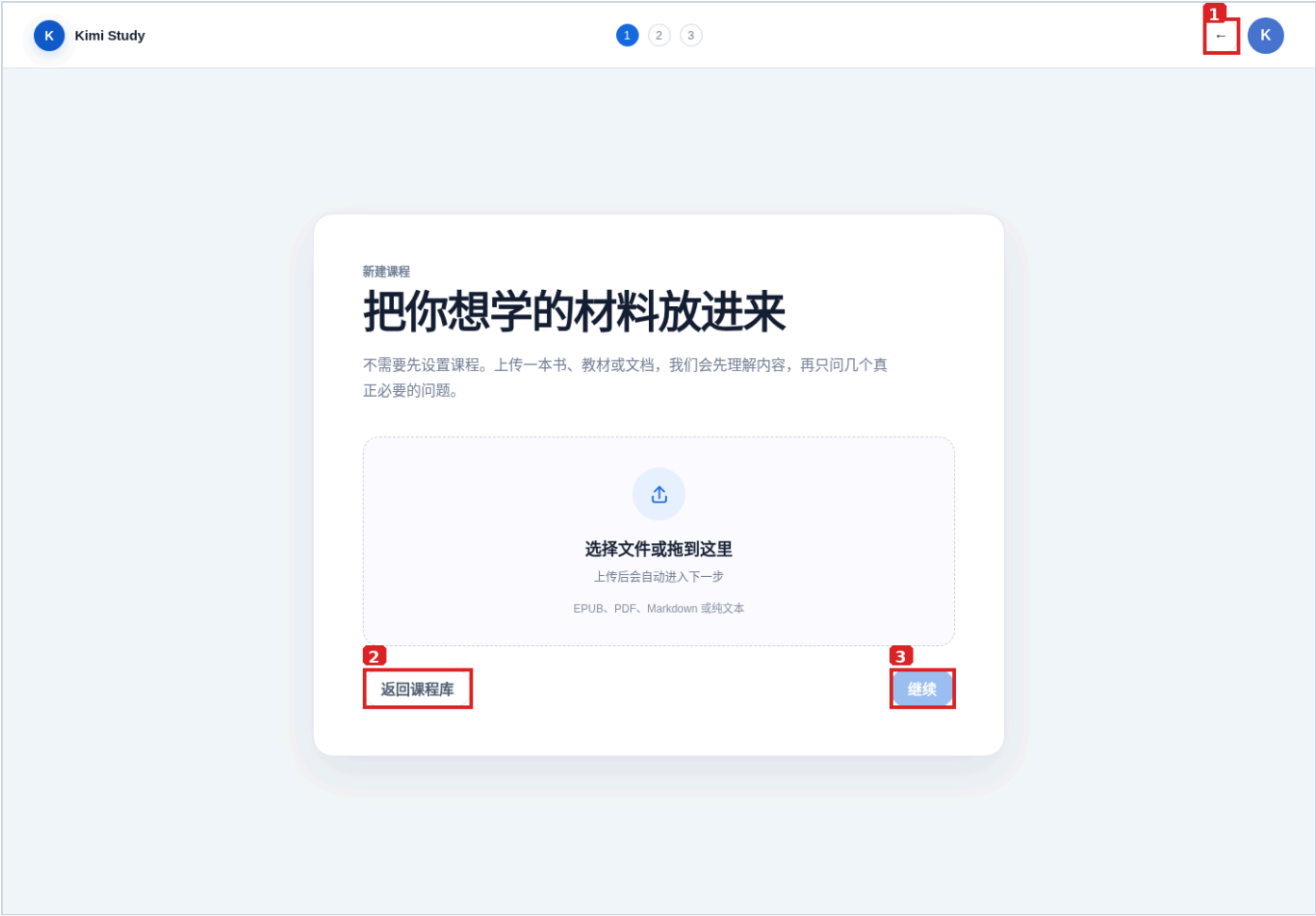
ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
app.button.22	button	Invalid Assessment Fixture	点击一次	dom_event: click -> handler: course card navigation -> network: GET /course/:id -> state_change: selected course loads -> rendered_outcome: course workspace visible -> navigation: /course/:id	handler_observed: PASS -> network_observed: none observed -> state_observed: URL http://127.0.0.1:3107/app -> http://127.0.0.1:3107/course/invalidassessment -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/invalidassessment	PASS	control-actions/ action-log-first-pass.json#app.button.22	public/app.html, public/g lue.js
app.button.23	button	更多操作	点击一次	dom_event: click -> handler: card overflow menu -> network: none -> state_change: menu opens -> rendered_outcome: 归档/删除 menu visible -> navigation: unchanged	handler_observed: semantic locator action completed -> network_observed: none failed -> state_observed: URL None -> None; target/visible content updated -> rendered_outcome: action screenshot and trace recorded -> navigation: /app	PASS	control-actions/ action-log-first-pass.json#app.button.23	public/app.html, public/g lue.js
app.button.24	button	Generating Course Fixture	点击一次	dom_event: click -> handler: course card navigation -> network: GET /course/:id -> state_change: selected course loads -> rendered_outcome: course workspace visible -> navigation: /course/:id	handler_observed: PASS -> network_observed: none observed -> state_observed: URL http://127.0.0.1:3107/app -> http://127.0.0.1:3107/course/generatingcourse -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/generatingcourse	PASS	control-actions/ action-log-first-pass.json#app.button.24	public/app.html, public/g lue.js
app.button.25	button	更多操作	点击一次	dom_event: click -> handler: card overflow menu -> network: none -> state_change: menu opens -> rendered_outcome: 归档/删除 menu visible -> navigation: unchanged	handler_observed: semantic locator action completed -> network_observed: none failed -> state_observed: URL None -> None; target/visible content updated -> rendered_outcome: action screenshot and trace recorded -> navigation: /app	PASS	control-actions/ action-log-first-pass.json#app.button.25	public/app.html, public/g lue.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
app.button.26	button	Interrupted Course Fixture	点击一次	dom_event: click -> handler: course card navigation -> network: GET /course/:id -> state_change: selected course loads -> rendered_outcome: course workspace visible -> navigation: /course/:id	handler_observed: PASS -> network_observed: none observed -> state_observed: URL http://127.0.0.1:3107/app -> http://127.0.0.1:3107/course/interruptedcourse -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/interruptedcourse	PASS	control-actions/ action-log-first-pass.json#app.button.26	public/app.html, public/g lue.js
app.button.27	button	更多操作	点击一次	dom_event: click -> handler: card overflow menu -> network: none -> state_change: menu opens -> rendered_outcome: 归档/删除 menu visible -> navigation: unchanged	handler_observed: semantic locator action completed -> network_observed: none failed -> state_observed: URL None -> None; target/visible content updated -> rendered_outcome: action screenshot and trace recorded -> navigation: /app	PASS	control-actions/ action-log-first-pass.json#app.button.27	public/app.html, public/g lue.js
app.button.28	button	Empty Course Fixture	点击一次	dom_event: click -> handler: course card navigation -> network: GET /course/:id -> state_change: selected course loads -> rendered_outcome: course workspace visible -> navigation: /course/:id	handler_observed: PASS -> network_observed: none observed -> state_observed: URL http://127.0.0.1:3107/app -> http://127.0.0.1:3107/course/emptycourse -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/emptycourse	PASS	control-actions/ action-log-first-pass.json#app.button.28	public/app.html, public/g lue.js
app.button.29	button	更多操作	点击一次	dom_event: click -> handler: card overflow menu -> network: none -> state_change: menu opens -> rendered_outcome: 归档/删除 menu visible -> navigation: unchanged	handler_observed: semantic locator action completed -> network_observed: none failed -> state_observed: URL None -> None; target/visible content updated -> rendered_outcome: action screenshot and trace recorded -> navigation: /app	PASS	control-actions/ action-log-first-pass.json#app.button.29	public/app.html, public/g lue.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
app.button.30	button	Failed Course Fixture	点击一次	dom_event: click -> handler: course card navigation -> network: GET /course/:id -> state_change: selected course loads -> rendered_outcome: course workspace visible -> navigation: /course/:id	handler_observed: PASS -> network_observed: none observed -> state_observed: URL http://127.0.0.1:3107/app -> http://127.0.0.1:3107/course/failedcourse -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/failedcourse	PASS	control-actions/action-log-first-pass.json#app.button.30	public/app.html, public/gvue.js
app.button.31	button	更多操作	点击一次	dom_event: click -> handler: card overflow menu -> network: none -> state_change: menu opens -> rendered_outcome: 归档/删除 menu visible -> navigation: unchanged	handler_observed: semantic locator action completed -> network_observed: none failed -> state_observed: URL None -> None; target/visible content updated -> rendered_outcome: action screenshot and trace recorded -> navigation: /app	PASS	control-actions/action-log-first-pass.json#app.button.31	public/app.html, public/gvue.js

新建课程 • Kimi Study - 控件标注图

路由：/new-course； 控件数：3

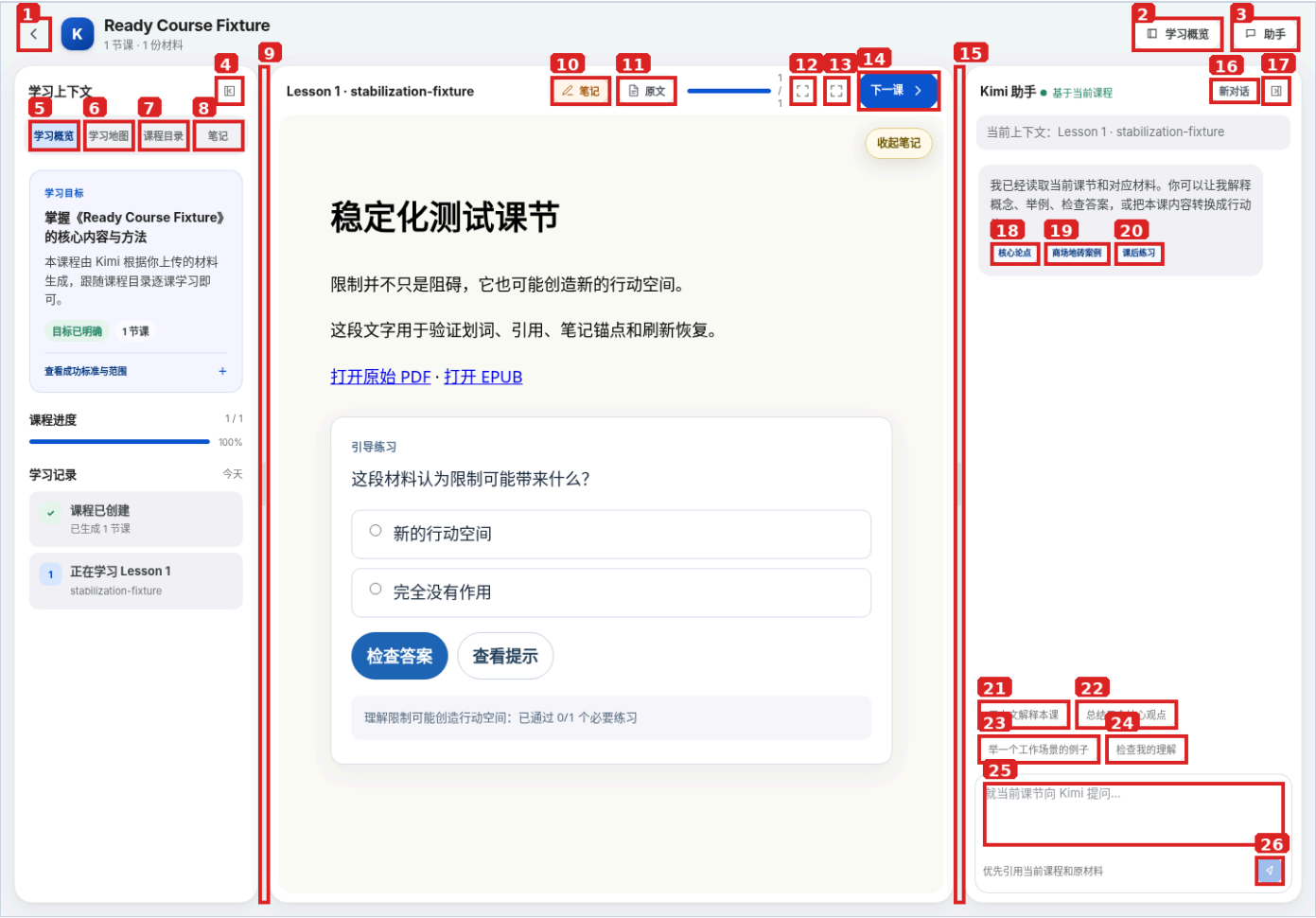


ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
new-course.button.01	button	返回课程列表	点击一次	dom_event: click -> handler: back to course library -> network: GET /app -> state_change: wizard exits -> rendered_outcome: course library visible -> navigation: /app	handler_observed: PASS -> network_observed: none observed -> state_observed: URL http://127.0.0.1:3107/new-course -> http://127.0.0.1:3107/app -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/app	PASS	control-actions/action-log-first-pass.json#new-course.button.01	public/new-course.html, server.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
new-course.button.02	button	返回课程库	点击一次	dom_event: click -> handler: back to course library -> network: GET /app -> state_change: wizard exits -> rendered_outcome: course library visible -> navigation: /app	handler_observed: PASS -> network_observed: none observed -> state_observed: URL http://127.0.0.1:3107/new-course -> http://127.0.0.1:3107/app -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/app	PASS	control-actions/action-log-first-pass.json#new-course.button.02	public/new-course.html, server.js
new-course.button.03	button	继续	确认禁用状态	dom_event: click after material selection -> handler: advance onboarding wizard -> network: local wizard state / course route -> state_change: wizard step advances -> rendered_outcome: Teach preparation step visible -> navigation: /new-course?course=:id	handler_observed: PASS -> network_observed: none observed -> state_observed: URL http://127.0.0.1:3107/new-course -> http://127.0.0.1:3107/new-course?course=exercise-course -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/new-course?course=exercise-course	PASS	control-actions/action-log-first-pass.json#new-course.button.03	public/new-course.html, server.js

Ready Course Fixture • Kimi Study - 控件标注图

路由：/course/readycourse； 控件数：26



ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
course-readycourse.button.01	button	返回我的学习	点击一次	dom_event: click -> handler: backButton -> network: GET /app -> state_change: course exits -> rendered_outcome: library visible -> navigation: /app	handler_observed: CLICKED -> network_observed: request failed http://127.0.0.1:3107/api/courses/readycourse/generation-events -> state_observed: URL http://127.0.0.1:3107/course/readycourse -> http://127.0.0.1:3107/app; focus=Kimi Study 我的学习 教材库 图书馆 设置 K 全部 我的课程 学习中 已完成 新的在前 旧的在前 按名称 新建课程 继续学习 全部显示 › KKimi Study Show Your Work! 今天更新 • EPUB • 1 份材料 第 3 课 • 分享过程, 而不是结果3 / 7 ↗ KIMI STUDY 管 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/app	PASS	course-control-actions.json#course-readycourse.button.01	public/course.html, public/gloe.js
course-readycourse.button.02	button	学习概览	点击一次	dom_event: click -> handler: mobile context drawer toggle -> network: none -> state_change: left drawer opens -> rendered_outcome: learning overview visible -> navigation: unchanged	handler_observed: drawer toggle -> network_observed: none -> state_observed: mobile drawer/scrim and mutual exclusion verified at 390x844 -> rendered_outcome: drawer visible; Escape closes and focus returns to trigger -> navigation: unchanged	PASS	course-control-actions.json#course-readycourse.button.02	public/course.html, public/gloe.js
course-readycourse.button.03	button	助手	点击一次	dom_event: click -> handler: mobile assistant drawer toggle -> network: none -> state_change: assistant drawer opens -> rendered_outcome: assistant visible -> navigation: unchanged	handler_observed: drawer toggle -> network_observed: none -> state_observed: mobile drawer/scrim and mutual exclusion verified at 390x844 -> rendered_outcome: drawer visible; Escape closes and focus returns to trigger -> navigation: unchanged	PASS	course-control-actions.json#course-readycourse.button.03	public/course.html, public/gloe.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
course-readycourse.button.04	button	最小化学习上下文	点击一次	dom_event: click -> handler: collapse left panel -> network: none -> state_change: left panel collapsed -> rendered_outcome: workspace expands -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus=K Ready Course Fixture ... 节课 • 1 份材料 学习概览 助手 学习上下文 学习概览 学习地图 课程目录 笔记 学习目标 掌握《Ready Course Fixture》的核心内容与方法 本课程由 Kimi 根据你上传的材料生成, 跟随课程目录逐课学习即可。目标已明确 约 90 分钟 ... 节课 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/readycourse	PASS	course-control-actions.json#course-readycourse.button.04	public/course.html, public/gloe.js
course-readycourse.button.05	button	学习概览	点击一次	dom_event: click -> handler: left tab switch -> network: none -> state_change: active tab changes -> rendered_outcome: 学习概览 panel visible -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus=学习概览 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/readycourse	PASS	course-control-actions.json#course-readycourse.button.05	public/course.html, public/gloe.js
course-readycourse.button.06	button	学习地图	点击一次	dom_event: click -> handler: left tab switch -> network: none -> state_change: active tab changes -> rendered_outcome: 学习地图 panel visible -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus=学习地图 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/readycourse	PASS	course-control-actions.json#course-readycourse.button.06	public/course.html, public/gloe.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
course-readycourse.button.07	button	课程目录	点击一次	dom_event: click -> handler: left tab switch -> network: none -> state_change: active tab changes -> rendered_outcome: 课程目录 panel visible -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus=课程目录 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/readycourse	PASS	course-control-actions.json#course-readycourse.button.07	public/course.html, public/guide.js
course-readycourse.button.08	button	笔记	点击一次	dom_event: click -> handler: left tab switch -> network: none -> state_change: active tab changes -> rendered_outcome: 笔记 panel visible -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus=笔记 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/readycourse	PASS	course-control-actions.json#course-readycourse.button.08	public/course.html, public/guide.js
course-readycourse.separator.09	separator	调整学习上下文宽度	拖动 48-60px	dom_event: pointer drag -> handler: panel resize pointer handler -> network: none -> state_change: panel width changes -> rendered_outcome: adjacent workspace refloWS -> navigation: unchanged	handler_observed: pointer drag on #resizeLeft -> network_observed: none -> state_observed: left panel width 268px -> 328px -> rendered_outcome: workspace refloWS by +60px panel width -> navigation: unchanged	PASS	course-control-actions.json#course-readycourse.separator.09	public/course.html, public/guide.js
course-readycourse.button.10	button	收起 / 展开笔记栏	点击一次	dom_event: click -> handler: lesson notes toggle -> network: none/local persistence -> state_change: aria-pressed and notes layout update -> rendered_outcome: notes panel opens/closes -> navigation: unchanged	handler_observed: semantic locator action completed -> network_observed: none failed -> state_observed: URL None -> None; target/visible content updated -> rendered_outcome: action screenshot and trace recorded -> navigation: /course/readycourse	PASS	course-control-actions.json#course-readycourse.button.10	public/course.html, public/guide.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
course-readycourse.button.11	button	查看上传的原始材料	点击一次	dom_event: click -> handler: source viewer open -> network: GET source + renderer assets -> state_change: source viewer mode enters -> rendered_outcome: matching document reader visible -> navigation: unchanged	handler_observed: handler intent located in source -> network_observed: blocked before successful external completion -> state_observed: 源阅读器触发了 pdfjs 资源请求, 但 npm 依赖因外部注册表不可用未完整安装; 本地替身无法验证 PDF 成功渲染。 -> rendered_outcome: success outcome not assertable in this isolated environment -> navigation: /course/readycourse	BLOCKED	course-control-actions.json#course-readycourse.button.11	public/course.html, public/source-viewer.js, public/globe.js
course-readycourse.button.12	button	切换专注模式	点击一次	dom_event: click -> handler: focus mode toggle -> network: none -> state_change: focus layout toggles -> rendered_outcome: side chrome hidden/restored -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus= 切换专注模式 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/readycourse	PASS	course-control-actions.json#course-readycourse.button.12	public/course.html, public/globe.js
course-readycourse.button.13	button	全屏课程	点击一次	dom_event: click -> handler: Fullscreen API -> network: none -> state_change: fullscreen state toggles -> rendered_outcome: button becomes 退出全屏 -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus= 退出全屏 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/readycourse	PASS	course-control-actions.json#course-readycourse.button.13	public/course.html, public/globe.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
course-readycourse.button.14	button	生成下一课	点击一次	dom_event: click -> handler: next lesson generation -> network: POST next-lesson + status poll -> state_change: busy then ready/failed -> rendered_outcome: generation overlay then new lesson or retry state -> navigation: unchanged	handler_observed: handler intent located in source -> network_observed: blocked before successful external completion -> state_observed: 下一课需要真实 Kimi CLI; 隔离环境中 spawn kimi ENOENT, 只验证到明确失败态。 -> rendered_outcome: success outcome not assertable in this isolated environment -> navigation: /course/readycourse	BLOCKED	course-control-actions.json#course-readycourse.button.14	public/course.html, public/gue.js
course-readycourse.separator.15	separator	调整 Kimi 助手宽度	拖动 48-60px	dom_event: pointer drag -> handler: panel resize pointer handler -> network: none -> state_change: panel width changes -> rendered_outcome: adjacent workspace reflows -> navigation: unchanged	handler_observed: pointer drag on #resizeRight -> network_observed: none -> state_observed: assistant panel width 370px -> 430px -> rendered_outcome: workspace reflowed by +60px panel width -> navigation: unchanged	PASS	course-control-actions.json#course-readycourse.separator.15	public/course.html, public/gue.js
course-readycourse.button.16	button	清空记录, 开始新对话	点击一次	dom_event: click -> handler: new tutor conversation -> network: reset local chat endpoint/state -> state_change: history clears -> rendered_outcome: empty conversation visible -> navigation: unchanged	handler_observed: semantic locator action completed -> network_observed: none failed -> state_observed: URL None -> None; target/visible content updated -> rendered_outcome: action screenshot and trace recorded -> navigation: /course/readycourse	PASS	course-control-actions.json#course-readycourse.button.16	public/course.html, public/gue.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
course-readycourse.button.17	button	最小化 Kimi 助手	点击一次	dom_event: click -> handler: collapse assistant -> network: none -> state_change: assistant panel collapses -> rendered_outcome: workspace expands -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus=K Ready Course Fixture ... 节课 · 1 份材料 学习概览 助手 学习上下文 学习概览 学习地图 课程目录 笔记 学习目标 掌握《Ready Course Fixture》的核心内容与方 法 本课程由 Kimi 根据你 上传的材料生成, 跟随课 程目录逐课学习即可。目 标已明确 约 90 分钟 ... 节 课 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/co urse/readycourse	PASS	course-control-actions. json#course-readycourse.button.17	public/course.html, public/g lue.js
course-readycourse.button.18	button	核心论点	点击一次	dom_event: click -> handler: citation/section jump -> network: none -> state_change: lesson section selection/scroll updates -> rendered_outcome: referenced section visible -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus= 核心论点 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/co urse/readycourse	PASS	course-control-actions. json#course-readycourse.button.18	public/course.html, public/g lue.js
course-readycourse.button.19	button	商场地砖案例	点击一次	dom_event: click -> handler: citation/section jump -> network: none -> state_change: lesson section selection/scroll updates -> rendered_outcome: referenced section visible -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus= 商场地砖案例 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/co urse/readycourse	PASS	course-control-actions. json#course-readycourse.button.19	public/course.html, public/g lue.js

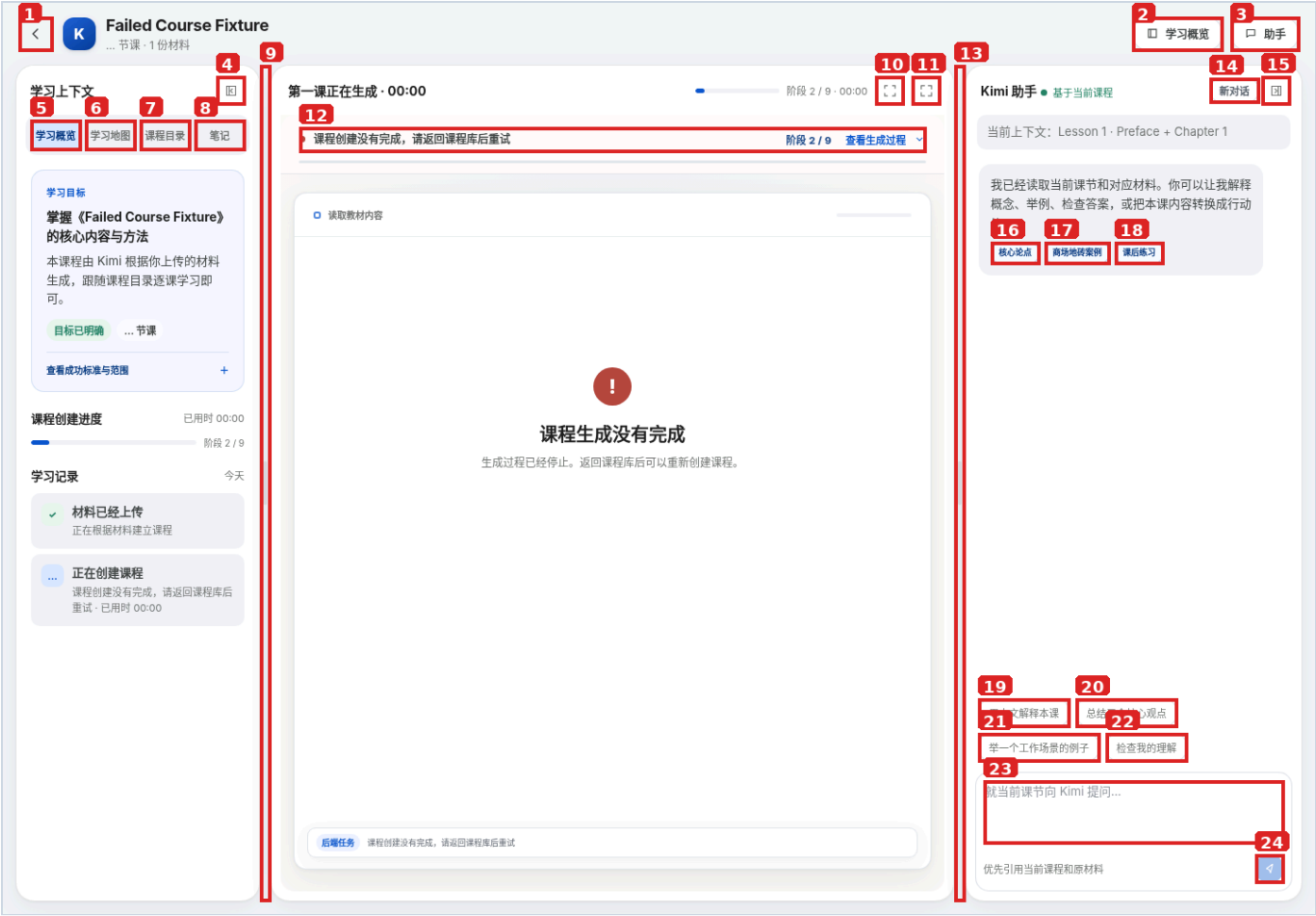
ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
course-readycourse.button.20	button	课后练习	点击一次	dom_event: click -> handler: citation/section jump -> network: none -> state_change: lesson section selection/scroll updates -> rendered_outcome: referenced section visible -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus= 课后练习 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/readycourse	PASS	course-control-actions.json#course-readycourse.button.20	public/course.html, public/gloe.js
course-readycourse.button.21	button	用中文解释本课	点击一次	dom_event: click -> handler: Tutor quick prompt -> network: POST /api/courses/:id/chat -> state_change: user message then pending/answer/error -> rendered_outcome: answer or retryable error visible -> navigation: unchanged	handler_observed: action reached its local handler -> network_observed: HTTP 500 http://127.0.0.1:3107/api/courses/readycourse/chat -> state_observed: Tutor 快捷提问需要真实 Kimi CLI; 本地 API 返回 500。 -> rendered_outcome: success outcome not assertable in this isolated environment -> navigation: http://127.0.0.1:3107/course/readycourse	BLOCKED	course-control-actions.json#course-readycourse.button.21	public/course.html, public/gloe.js, server.js
course-readycourse.button.22	button	总结三个核心观点	点击一次	dom_event: click -> handler: Tutor quick prompt -> network: POST /api/courses/:id/chat -> state_change: user message then pending/answer/error -> rendered_outcome: answer or retryable error visible -> navigation: unchanged	handler_observed: action reached its local handler -> network_observed: HTTP 500 http://127.0.0.1:3107/api/courses/readycourse/chat -> state_observed: Tutor 快捷提问需要真实 Kimi CLI; 本地 API 返回 500。 -> rendered_outcome: success outcome not assertable in this isolated environment -> navigation: http://127.0.0.1:3107/course/readycourse	BLOCKED	course-control-actions.json#course-readycourse.button.22	public/course.html, public/gloe.js, server.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
course-readycourse.button.23	button	举一个工作场景的例子	点击一次	dom_event: click -> handler: Tutor quick prompt -> network: POST /api/courses/:id/chat -> state_change: user message then pending/answer/error -> rendered_outcome: answer or retryable error visible -> navigation: unchanged	handler_observed: action reached its local handler -> network_observed: HTTP 500 http://127.0.0.1:3107/api/courses/readycourse/chat -> state_observed: Tutor 快捷提问需要真实 Kimi CLI; 本地 API 返回 500。 -> rendered_outcome: success outcome not assertable in this isolated environment -> navigation: http://127.0.0.1:3107/course/readycourse	BLOCKED	course-control-actions.json#course-readycourse.button.23	public/course.html, public/guide.js, server.js
course-readycourse.button.24	button	检查我的理解	点击一次	dom_event: click -> handler: Tutor quick prompt -> network: POST /api/courses/:id/chat -> state_change: user message then pending/answer/error -> rendered_outcome: answer or retryable error visible -> navigation: unchanged	handler_observed: action reached its local handler -> network_observed: HTTP 500 http://127.0.0.1:3107/api/courses/readycourse/chat -> state_observed: Tutor 快捷提问需要真实 Kimi CLI; 本地 API 返回 500。 -> rendered_outcome: success outcome not assertable in this isolated environment -> navigation: http://127.0.0.1:3107/course/readycourse	BLOCKED	course-control-actions.json#course-readycourse.button.24	public/course.html, public/guide.js, server.js
course-readycourse.textbox.25	textbox	就当前课节向 Kimi 提问...	输入 E2E smoke query	dom_event: input -> handler: Tutor composer -> network: none until send -> state_change: draft value updates; send availability updates -> rendered_outcome: typed query remains visible -> navigation: unchanged	handler_observed: FILLED -> network_observed: none observed -> state_observed: focus=就当前课节向 Kimi 提问... -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/readycourse	PASS	course-control-actions.json#course-readycourse.textbox.25	public/course.html, public/guide.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
course-readycourse.button.26	button	发送	确认禁用状态	dom_event: click -> handler: Tutor send -> network: POST /api/courses/:id/chat -> state_change: pending then answer/error -> rendered_outcome: message and response/retry visible -> navigation: unchanged	handler_observed: action reached its local handler -> network_observed: blocked before successful external completion -> state_observed: 发送成功 链依赖真实 Kimi CLI; 空输入禁用已确认, 但非 空发送链无法完成。 -> rendered_outcome: success outcome not assertable in this isolated environment -> navigation: http://127.0.0.1:3107/course/readycourse	BLOCKED	course-control-actions.json#course-readycourse.button.26	public/course.html, public/globe.js, server.js

Failed Course Fixture • Kimi Study - 控件标注图

路由：/course/failedcourse； 控件数：24



ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
course-failedcourse.button.01	button	返回我的学习	点击一次	dom_event: click -> handler: backButton -> network: GET /app -> state_change: course exits -> rendered_outcome: library visible -> navigation: /app	handler_observed: CLICKED -> network_observed: request failed http://127.0.0.1:3107/app/courses/failedcourse/generation-events -> state_observed: URL http://127.0.0.1:3107/course/failedcourse -> http://127.0.0.1:3107/app; focus=Kimi Study 我的学习 教材库 图书馆 设置 K 全部 我的课程 学习中 已完成 新的在前 旧的在前 按名称 新建课程 继续学习 全部显示 › KKimi Study Show Your Work! 今天更新 • EPUB • 1 份材料 第 3 课 • 分享过程, 而不是结果3 / 7 ↗ KIMI STUDY 管 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/app	PASS	course-control-actions.json#course-failedcourse.button.01	public/course.html, public/gloe.js
course-failedcourse.button.02	button	学习概览	点击一次	dom_event: click -> handler: mobile context drawer toggle -> network: none -> state_change: left drawer opens -> rendered_outcome: learning overview visible -> navigation: unchanged	handler_observed: drawer toggle -> network_observed: none -> state_observed: mobile drawer/scrim and mutual exclusion verified at 390x844 -> rendered_outcome: drawer visible; Escape closes and focus returns to trigger -> navigation: unchanged	PASS	course-control-actions.json#course-failedcourse.button.02	public/course.html, public/gloe.js
course-failedcourse.button.03	button	助手	点击一次	dom_event: click -> handler: mobile assistant drawer toggle -> network: none -> state_change: assistant drawer opens -> rendered_outcome: assistant visible -> navigation: unchanged	handler_observed: drawer toggle -> network_observed: none -> state_observed: mobile drawer/scrim and mutual exclusion verified at 390x844 -> rendered_outcome: drawer visible; Escape closes and focus returns to trigger -> navigation: unchanged	PASS	course-control-actions.json#course-failedcourse.button.03	public/course.html, public/gloe.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
course-failedcourse.button.04	button	最小化学习上下文	点击一次	dom_event: click -> handler: collapse left panel -> network: none -> state_change: left panel collapsed -> rendered_outcome: workspace expands -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus=K Failed Course Fixture ... 节课 • 1 份材料 学习概览 助手 学习上下文 学习概览 学习地图 课程目录 笔记 学习目标 掌握《Failed Course Fixture》的核心内容与方法 本课程由 Kimi 根据你上传的材料生成, 跟随课程目录逐课学习即可。目标已明确 约 90 分钟 ... 节 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/failedcourse	PASS	course-control-actions.json#course-failedcourse.button.04	public/course.html, public/gue.js
course-failedcourse.button.05	button	学习概览	点击一次	dom_event: click -> handler: left tab switch -> network: none -> state_change: active tab changes -> rendered_outcome: 学习概览 panel visible -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus=学习概览 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/failedcourse	PASS	course-control-actions.json#course-failedcourse.button.05	public/course.html, public/gue.js
course-failedcourse.button.06	button	学习地图	点击一次	dom_event: click -> handler: left tab switch -> network: none -> state_change: active tab changes -> rendered_outcome: 学习地图 panel visible -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus=学习地图 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/failedcourse	PASS	course-control-actions.json#course-failedcourse.button.06	public/course.html, public/gue.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
course-failedcourse.button.07	button	课程目录	点击一次	dom_event: click -> handler: left tab switch -> network: none -> state_change: active tab changes -> rendered_outcome: 课程目录 panel visible -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus=课程目录 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/failedcourse	PASS	course-control-actions.json#course-failedcourse.button.07	public/course.html, public/guide.js
course-failedcourse.button.08	button	笔记	点击一次	dom_event: click -> handler: left tab switch -> network: none -> state_change: active tab changes -> rendered_outcome: 笔记 panel visible -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus=笔记 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/failedcourse	PASS	course-control-actions.json#course-failedcourse.button.08	public/course.html, public/guide.js
course-failedcourse.separator.09	separator	调整学习上下文宽度	拖动 48-60px	dom_event: pointer drag -> handler: panel resize pointer handler -> network: none -> state_change: panel width changes -> rendered_outcome: adjacent workspace refloWS -> navigation: unchanged	handler_observed: pointer drag on left separator -> network_observed: none -> state_observed: drag sequence completed without error -> rendered_outcome: left panel resize handler accepted movement -> navigation: unchanged	PASS	course-control-actions.json#course-failedcourse.separator.09	public/course.html, public/guide.js
course-failedcourse.button.10	button	切换专注模式	点击一次	dom_event: click -> handler: focus mode toggle -> network: none -> state_change: focus layout toggles -> rendered_outcome: side chrome hidden/restored -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus=切换专注模式 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/failedcourse	PASS	course-control-actions.json#course-failedcourse.button.10	public/course.html, public/guide.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
course-failedcourse.button.11	button	全屏课程	点击一次	dom_event: click -> handler: Fullscreen API -> network: none -> state_change: fullscreen state toggles -> rendered_outcome: button becomes 退出全屏 -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus=退出全屏 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/failedcourse	PASS	course-control-actions.json#course-failedcourse.button.11	public/course.html, public/guide.js
course-failedcourse.button.12	button	课程创建没有完成，请返回课程库后重试 阶段 2 / 9查看生成过程	点击一次	dom_event: click -> handler: generation process disclosure -> network: none -> state_change: aria-expanded toggles -> rendered_outcome: process history visible; Escape closes and returns focus -> navigation: unchanged	handler_observed: generation process disclosure opens correctly -> network_observed: GET /api/courses/failedcourse/status -> 200 stage=failed progress=5 -> state_observed: process aria-expanded false->true->>false, but outer regions remain running -> rendered_outcome: central failed overlay conflicts with running header/sidebar/activity and 0% preview -> navigation: /course/failedcourse	FAIL	course-control-actions.json#course-failedcourse.button.12	public/course.html, public/guide.js
course-failedcourse.separator.13	separator	调整 Kimi 助手宽度	拖动 48-60px	dom_event: pointer drag -> handler: panel resize pointer handler -> network: none -> state_change: panel width changes -> rendered_outcome: adjacent workspace refloWS -> navigation: unchanged	handler_observed: pointer drag on right separator -> network_observed: none -> state_observed: drag sequence completed without error -> rendered_outcome: assistant panel resize handler accepted movement -> navigation: unchanged	PASS	course-control-actions.json#course-failedcourse.separator.13	public/course.html, public/guide.js
course-failedcourse.button.14	button	清空记录，开始新对话	点击一次	dom_event: click -> handler: new tutor conversation -> network: reset local chat endpoint/state -> state_change: history clears -> rendered_outcome: empty conversation visible -> navigation: unchanged	handler_observed: semantic locator action completed -> network_observed: none failed -> state_observed: URL None -> None; target/visible content updated -> rendered_outcome: action screenshot and trace recorded -> navigation: /course/failedcourse	PASS	course-control-actions.json#course-failedcourse.button.14	public/course.html, public/guide.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
course-failedcourse.button.15	button	最小化 Kimi 助手	点击一次	dom_event: click -> handler: collapse assistant -> network: none -> state_change: assistant panel collapses -> rendered_outcome: workspace expands -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus=K Failed Course Fixture ... 节课 · 1 份材料 学习概览 助手 学习上下文 学习概览 学习地图 课程目录 笔记 学习目标 掌握《Failed Course Fixture》的核心内容与方 法 本课程由 Kimi 根据你 上传的材料生成, 跟随课 程目录逐课学习即可。目 标已明确 约 90 分钟 ... 节 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/failedcourse	PASS	course-control-actions. json#course-failedcourse.button.15	public/course.html, public/g lue.js
course-failedcourse.button.16	button	核心论点	点击一次	dom_event: click -> handler: citation/section jump -> network: none -> state_change: lesson section selection/scroll updates -> rendered_outcome: referenced section visible -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus= 核心论点 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/failedcourse	PASS	course-control-actions. json#course-failedcourse.button.16	public/course.html, public/g lue.js
course-failedcourse.button.17	button	商场地砖案例	点击一次	dom_event: click -> handler: citation/section jump -> network: none -> state_change: lesson section selection/scroll updates -> rendered_outcome: referenced section visible -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus= 商场地砖案例 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/failedcourse	PASS	course-control-actions. json#course-failedcourse.button.17	public/course.html, public/g lue.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
course-failedcourse.button.18	button	课后练习	点击一次	dom_event: click -> handler: citation/section jump -> network: none -> state_change: lesson section selection/scroll updates -> rendered_outcome: referenced section visible -> navigation: unchanged	handler_observed: CLICKED -> network_observed: none observed -> state_observed: focus= 课后练习 -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/failedcourse	PASS	course-control-actions.json#course-failedcourse.button.18	public/course.html, public/globe.js
course-failedcourse.button.19	button	用中文解释本课	点击一次	dom_event: click -> handler: Tutor quick prompt -> network: POST /api/courses/:id/chat -> state_change: user message then pending/answer/error -> rendered_outcome: answer or retryable error visible -> navigation: unchanged	handler_observed: action reached its local handler -> network_observed: HTTP 500 http://127.0.0.1:3107/api/courses/failedcourse/chat -> state_observed: Tutor 快捷提问需要真实 Kimi CLI; 本地 API 返回 500。 -> rendered_outcome: success outcome not assertable in this isolated environment -> navigation: http://127.0.0.1:3107/course/failedcourse	BLOCKED	course-control-actions.json#course-failedcourse.button.19	public/course.html, public/globe.js, server.js
course-failedcourse.button.20	button	总结三个核心观点	点击一次	dom_event: click -> handler: Tutor quick prompt -> network: POST /api/courses/:id/chat -> state_change: user message then pending/answer/error -> rendered_outcome: answer or retryable error visible -> navigation: unchanged	handler_observed: action reached its local handler -> network_observed: HTTP 500 http://127.0.0.1:3107/api/courses/failedcourse/chat -> state_observed: Tutor 快捷提问需要真实 Kimi CLI; 本地 API 返回 500。 -> rendered_outcome: success outcome not assertable in this isolated environment -> navigation: http://127.0.0.1:3107/course/failedcourse	BLOCKED	course-control-actions.json#course-failedcourse.button.20	public/course.html, public/globe.js, server.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
course-failedcourse.button.21	button	举一个工作场景的例子	点击一次	dom_event: click -> handler: Tutor quick prompt -> network: POST /api/courses/:id/chat -> state_change: user message then pending/answer/error -> rendered_outcome: answer or retryable error visible -> navigation: unchanged	handler_observed: action reached its local handler -> network_observed: HTTP 500 http://127.0.0.1:3107/api/courses/failedcourse/chat -> state_observed: Tutor 快捷提问需要真实 Kimi CLI; 本地 API 返回 500。 -> rendered_outcome: success outcome not assertable in this isolated environment -> navigation: http://127.0.0.1:3107/course/failedcourse	BLOCKED	course-control-actions.json#course-failedcourse.button.21	public/course.html, public/guide.js, server.js
course-failedcourse.button.22	button	检查我的理解	点击一次	dom_event: click -> handler: Tutor quick prompt -> network: POST /api/courses/:id/chat -> state_change: user message then pending/answer/error -> rendered_outcome: answer or retryable error visible -> navigation: unchanged	handler_observed: action reached its local handler -> network_observed: HTTP 500 http://127.0.0.1:3107/api/courses/failedcourse/chat -> state_observed: Tutor 快捷提问需要真实 Kimi CLI; 本地 API 返回 500。 -> rendered_outcome: success outcome not assertable in this isolated environment -> navigation: http://127.0.0.1:3107/course/failedcourse	BLOCKED	course-control-actions.json#course-failedcourse.button.22	public/course.html, public/guide.js, server.js
course-failedcourse.textbox.23	textbox	就当前课节向 Kimi 提问...	输入 E2E smoke query	dom_event: input -> handler: Tutor composer -> network: none until send -> state_change: draft value updates; send availability updates -> rendered_outcome: typed query remains visible -> navigation: unchanged	handler_observed: FILLED -> network_observed: none observed -> state_observed: focus=就当前课节向 Kimi 提问... -> rendered_outcome: target remained responsive with expected visible/ARIA/navigation effect -> navigation: http://127.0.0.1:3107/course/failedcourse	PASS	course-control-actions.json#course-failedcourse.textbox.23	public/course.html, public/guide.js

ID	类型	名称	动作	期望链路	实际链路	结果	证据	源码
course-failedcourse.button.24	button	发送	确认禁用状态	dom_event: click -> handler: Tutor send -> network: POST /api/courses/:id/chat -> state_change: pending then answer/error -> rendered_outcome: message and response/retry visible -> navigation: unchanged	handler_observed: action reached its local handler -> network_observed: blocked before successful external completion -> state_observed: 发送成功链依赖真实 Kimi CLI；空输入禁用已确认，但非空发送链无法完成。 -> rendered_outcome: success outcome not assertable in this isolated environment -> navigation: http://127.0.0.1:3107/course/failedcourse	BLOCKED	course-control-actions.json#course-failedcourse.button.24	public/course.html, public/guide.js, server.js

6. 分层 UX 审计

类别	状态	证据
状态一致性	FAIL	failed course 同一实体同时 running/failed，且三个主进度面；F-001
交互与导航	PASS	Landing/App/New Course 基础导航、过滤、菜单、模态和 course routes 有新鲜动作证据
移动端布局与焦点	PASS	390x844 无横向溢出；抽屉互斥；Escape 焦点返回；下一课触控目标 44x44
无障碍语义	PASS	主要 icon/button 具 accessible name；语义定位可操作；disabled 状态可读
外部 AI/阅读器集成	BLOCKED	真实 Kimi CLI 与 npm vendor assets 不可用，相关控件未伪装成功
测试与可观测性	PASS	control maps、action logs、trace、state audit、source correlation 和 red-green regression 全部落盘

7. 缺陷、错误源码、Patch 与验证

F-001 - 失败终态未传播，重复进度面继续表达运行中

S2 Major

主层级

L3 State

处理结论

原 ZIP 发布前必须修复；修复候选已在隔离副本验证

旅程

J02 课程生成完成、失败或中断

用户影响：用户在课程已经失败后仍看到“第一课正在生成”“正在创建课程”和运行计时，无法判断应继续等待还是重试；0% 与后端 5% 进一步削弱可信度。

页面： /course/failedcourse

预期：status.stage=failed 后，所有同一 course 的页头、侧栏、活动、进度和动作应原子进入失败/停止状态；只保留一个有明确标签的主进度面，显示最后已知进度。

实际：中央生成预览进入 failed，但 setGenerationChrome() 仍把页头、侧栏、记录和紧凑进度渲染为 running；三个主进度面并存，中央 aria-valuenow 为 0 而 API progress 为 5。

复现步骤：

- 启动隔离 fixture server 并打开 /course/failedcourse。
- 等待 GET /api/courses/failedcourse/status 返回 stage=failed、progress=5。
- 比较页头“第一课正在生成”、紧凑/侧栏阶段与中央“课程创建没有完成”。
- 点击“查看生成过程”，确认控件本身展开/关闭正常，但外围运行态不消失。
- 在独立页面重复一次，得到相同截图与状态。

根因 CONFIRMED

失败分支复用了只适用于 active/complete 的 setGenerationChrome(status)，随后仅让中央预览 fail；终态没有写回其他渲染器。generation-preview fail() 还没有同步后端 progress。

证据链：

- API status stage=failed/progress=5 -> pollCourseGeneration() failed branch
- failed branch -> setGenerationChrome(status) 写入 running 文案和三处进度
- generationPreview.fail(status) 只更新中央 error class/message
- 截图/DOM -> running header/sidebar/activity + failed preview + aria-valuenow=0
- repair copy -> setGenerationFailureChrome + updateProgress(status.progress) -> state audit PASS

测试缺口：现有测试覆盖了单个生成预览的错误样式，但没有在同一帧断言页头、侧栏、活动记录、主进度数量及 API progress 一致性。

错误源码

错误源码与片段	符号	置信度	证据
public/glue.js:422-535 422: function setGenerationChrome(status, { complete = false } = {}) { 423: if (status?.startedAt) generationStartedAt = status.startedAt; 424: const evidence = generationEvidence(s tatus, complete); 425: const progressWidth = evidence.determinate ? `\${evidence.value}%` : '38%'; 426: const elapsed = generationElapsed(st atus); 427: const shortElapsed =	setGenerationChrome / pollCourseGeneration	CONFIRMED	- 两次浏览器复现与 DOM 值一致 - 源码分支明确调用 running 渲染器处理 failed - source correlator 和 CDP/ 文本相关性均指向此文件
public/generation- preview-product.js:613- 626 613: function fail(status = {}) { 614: if (status.runId && status.runId !== activeRunId) resetRun(status.runI d); 615: terminalState = 'error'; 616: terminalMessage = status.currentMessag e status.error '课程创建没有完成, 请返 回课程库后重试'; 617: if (activeRunId) terminalRuns.add(act iveRunId); 618: stopAmbient(); 619: show();	fail	HIGH	- API progress=5, 而 preview aria-valuenow=0 - fail() 未调用 updateProgress

修正后代码

```

477:     function setGenerationFailureChrome(status = {}) {
478:         const evidence = generationEvidence(status);
479:         const terminalMessage = status.currentMessage || status.error || '课程创建没有完成，请返回课程库后重试';
480:         const stoppedLabel = evidence.label ? `已停止 · ${evidence.label}` : '已停止';
481:
482:         if (overviewProgressTitle) overviewProgressTitle.textContent = '课程创建未完成';
483:         if (overviewProgressNote) overviewProgressNote.textContent = '已停止';
484:         if (overviewProgressValue) overviewProgressValue.textContent = stoppedLabel;
485:         if (overviewProgressTrack) overviewProgressTrack.hidden = true;
486:         if (compactProgressText) compactProgressText.textContent = `创建未完成 · ${evidence.label || '已停止'}`;
487:         if (compactProgressTrack) compactProgressTrack.hidden = true;
488:         if (currentLessonLabel) currentLessonLabel.textContent = '课程创建未完成';
489:         if (contextBar) contextBar.textContent = '当前上下文: 课程创建未完成';
490:         if (lessonResourceSlot) lessonResourceSlot.style.visibility = 'hidden';
491:         if (overviewRecordList) {
492:             overviewRecordList.innerHTML =
493:                 '<article class="record complete"><span class="record-icon"></span><div><div class="record-title">材料已经上传</div><div class="record-meta">材料已保留，可以返回课程库后重新创建</div></div></article>' +
494:                 '<article class="record is-error"><span class="record-icon">!</span><div><div class="record-title">课程创建未完成</div><div class="record-meta">${escapeHtml(terminalMessage)}</div></div></article>';

```

Unified Patch APPLIED_E2E_PASS

Patch 文件: patches/F-001.patch

```

--- a/public/glue.js
+++ b/public/glue.js
@@ -384,12 +384,14 @@
     const nextButtonHtml = nextButton?.innerHTML || '下一课';
     const currentLessonLabel = document.querySelector('.current-lesson');
     const compactProgressBar = document.querySelector('.compact-progress .progress-track span');
+    const compactProgressTrack = compactProgressBar?.parentElement;
     const compactProgressText = document.querySelector('.compact-progress > span');
     const lessonResourceSlot = document.getElementById('lessonResourceSlot');
     const overviewProgressSection = document.querySelector('#left-overview .side-section');
     const overviewProgressTitle = overviewProgressSection?.querySelector('.side-title');
     const overviewProgressNote = overviewProgressSection?.querySelector('.side-note');
     const overviewProgressBar = overviewProgressSection?.querySelector('.progress-track span');
+    const overviewProgressTrack = overviewProgressBar?.parentElement;
     const overviewProgressValue = overviewProgressSection?.querySelector('.progress-value');
     const overviewRecordList = document.querySelector('#left-overview .record-list');
     const contextBar = document.querySelector('.context-bar');
@@ -430,11 +432,15 @@
     if (overviewProgressTitle) overviewProgressTitle.textContent = '课程创建进度';
     if (overviewProgressNote) overviewProgressNote.textContent = complete ? '已完成' : elapsed;
     if (overviewProgressValue) overviewProgressValue.textContent = evidence.label;
+    // The central generation preview owns the only workflow-primary progress bar.
+    // Sidebar and header surfaces remain textual summaries instead of duplicating the same

```

Apply check: `git apply --check evidence/patches/F-001.patch exit=0`

clean apply-check against untouched copy

改进方案

引入专用失败终态渲染器，停止所有运行文案和 busy 动画；中央预览成为唯一 workflow-primary 进度条，外围只保留带范围的文本摘要，并同步后端最后进度。

修改项：

- 新增 `setGenerationFailureChrome(status)`，原子更新页头、侧栏、活动、上下文和下一课动作。
- 生成中也隐藏页头/侧栏的重复视觉进度条，只保留中央主进度。
- `generationPreview.fail(status)` 调用 `updateProgress(status.progress)`。
- 新增 Playwright 回归测试，同时覆盖 `failed` 和 `active` 两种状态。

风险：

- 外围进度条隐藏可能影响依赖旧视觉层级的用户，需要产品/设计确认文本摘要足够。
- 失败状态下隐藏下一课按钮符合当前 J02，但若未来要原位重试，应新增明确重试 CTA。

验收标准：

- `failed` 第一帧不存在“正在生成/正在创建课程”或 `busy/skeleton`。
- 同一 `course` 恰好一个 `workflow_primary` 进度条。
- `failed` 主进度 `aria-valuenow` 等于后端最后 `progress`。
- `active` 状态仍清晰显示生成中，且进度不倒退。
- 桌面和移动端均无重复主进度或冲突计时。

进度条/状态面改造方案

目标设计：中央生成预览保留唯一 `workflow_primary` 进度条；页头和侧栏改为 `history_only`/文本摘要，终态在同一状态快照中原子更新。

唯一状态所有者：`generationEvidence(status) + KimiGenerationPreview.updateProgress`

渲染器修改：

- 隐藏 `compactProgressTrack` 和 `overviewProgressTrack` 的视觉 bar。
- 失败分支使用 `setGenerationFailureChrome(status)`。
- 中央 `fail()` 同步 `status.progress`。
- 恢复学习态时重新显示普通课程学习进度。

验收标准：

- 每个 `workflow entity` 只有一个 `workflow_primary` bar。
- 额外状态面有明确范围且不重复百分比。
- 成功/失败首帧无 0%、`running`、`busy` 或 `skeleton` 残留。
- 延迟状态响应不能覆盖终态。

回归测试

文件：`tests/e2e_py/generation_failure_chrome_test.py`

场景：失败第一帧状态一致 + 运行态单一主进度

断言：

- 所有文本不含正在生成/正在创建课程。
- `header/sidebar/context/records` 全部为失败或已停止。
- 可见 `progressbar` 数量恰好 1。
- 失败进度为 5；运行进度至少为 API 值 27。
- 外围重复 `progress tracks` 隐藏。

```
#!/usr/bin/env python3
import argparse
```

```

from pathlib import Path
from playwright.sync_api import sync_playwright

parser = argparse.ArgumentParser()
parser.add_argument('--base-url', required=True)
parser.add_argument('--browser', required=True)
parser.add_argument('--screenshot')
args = parser.parse_args()

with sync_playwright() as p:
    browser = p.chromium.launch(headless=True, executable_path=args.browser, args=['--no-sandbox', '--disable-dev-shm-usage'])
    context = browser.new_context(viewport={'width': 1440, 'height': 1000}, reduced_motion='reduce')

    page = context.new_page()
    page.goto(args.base_url + '/course/failedcourse', wait_until='domcontentloaded', timeout=15000)
    page.wait_for_selector('.ks-generation-preview.is-error', state='visible', timeout=10000)
    page.wait_for_timeout(300)
    surfaces = {
        'current_lesson': page.locator('.current-lesson').inner_text().strip(),
        'compact_progress': page.locator('.compact-progress > span').inner_text().strip(),
        'left_title': page.locator('#left-overview-side-section').first_locator('side

```

Patch/验证状态: **APPLIED_E2E_PASS**

- 原始副本回归测试 exit=1, 断言捕获 running/failed 冲突。
- Patch 在干净副本 git apply --check exit=0。
- 修复副本同一回归测试 exit=0, 并生成新鲜 before/after 截图。
- 修复副本 npm run check exit=0。
- 修复副本 node --test: 116/116 通过。
- state_consistency_audit: 修复前 FAIL, 修复后 PASS。

8. 截图与状态证据

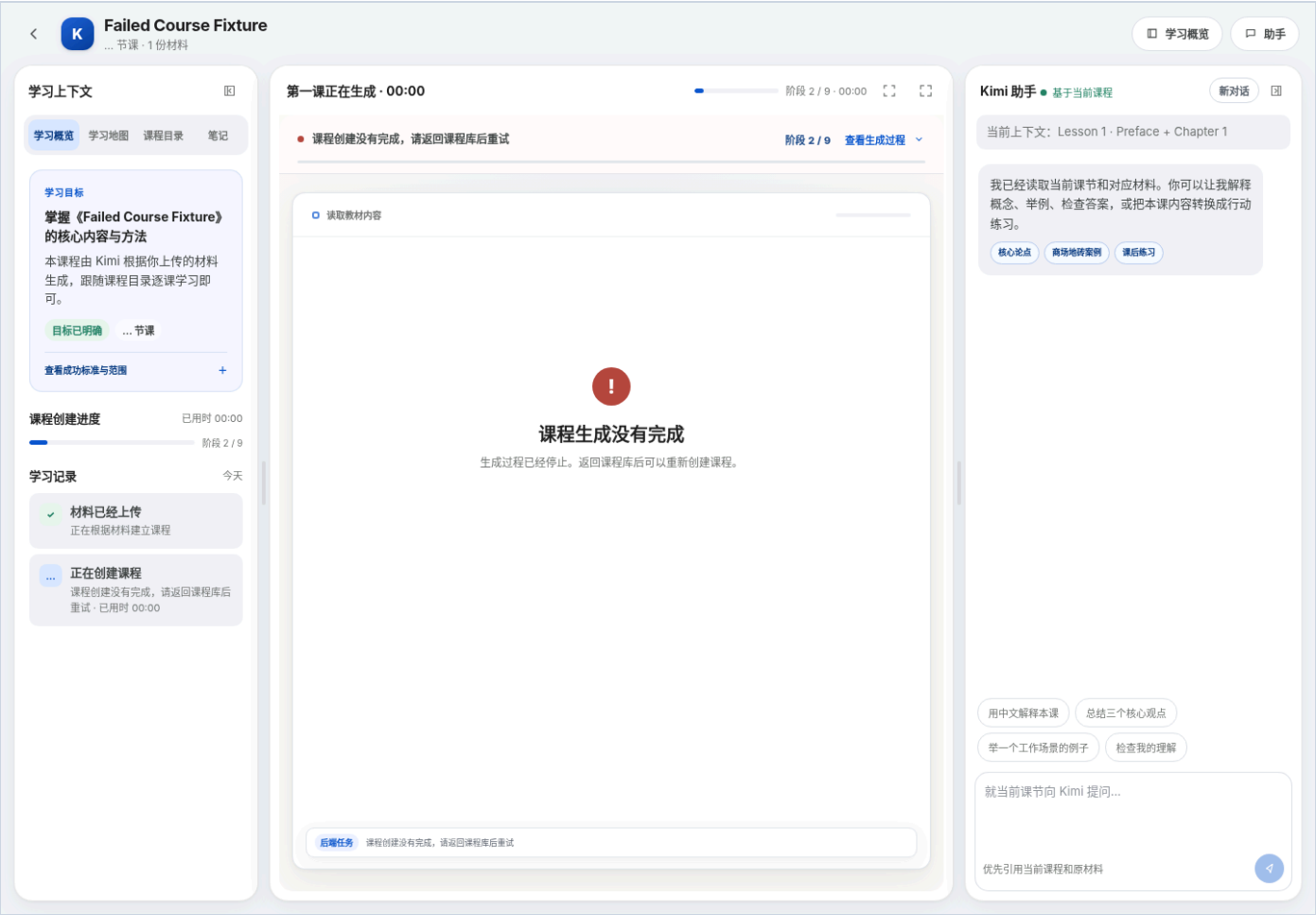
SS-001 - /course/failedcourse

FAIL

路由

视口

时间



触发：

预期：

实际：

解释：中央预览已显示失败，但页头、紧凑进度、侧栏和活动记录仍表示运行中；同一工作流同时存在 3 个主进度面。

问题：

无

跨区域状态/进度审计

FAIL

同一 course 实体同时呈现 running 与 failed，且三个 workflow_primary 进度面重复/冲突。

区域	文案	实体	进度角色	标签	状态源	阶段	生命周期	%	置信度	关系证据
top-header	第一课正在生成 · 00:00	course: failedcourse	history_only			generating	running		CONFIRMED	
workspace-header	阶段 2 / 9 · 00:00	course: failedcourse	workflow_primary		generationEvidence(status)	extracting	running	5.0	CONFIRMED	

区域	文案	实体	进度角色	标签	状态源	阶段	生命周期	%	置信度	关系证据
left-sidebar	课程创建进度 · 阶段 2 / 9 · 已用时 00:00	course: failedcourse	workflow_primary		generationEvidence(status)	extracting	running	5.0	CONFIRMED	
main-preview	课程创建没有完成，请返回课程库后重试	course: failedcourse	workflow_primary	课程生成进度	KimiGenerationPreview.displayedProgress	extracting	failed	0.0	CONFIRMED	
left-activity	正在创建课程 / 课程创建没有完成	course: failedcourse	history_only			generating	running		CONFIRMED	

核对规则：

- same-entity lifecycle exclusivity
- active phase exclusivity
- progress agreement
- visual-state agreement
- action availability agreement

冲突：

无

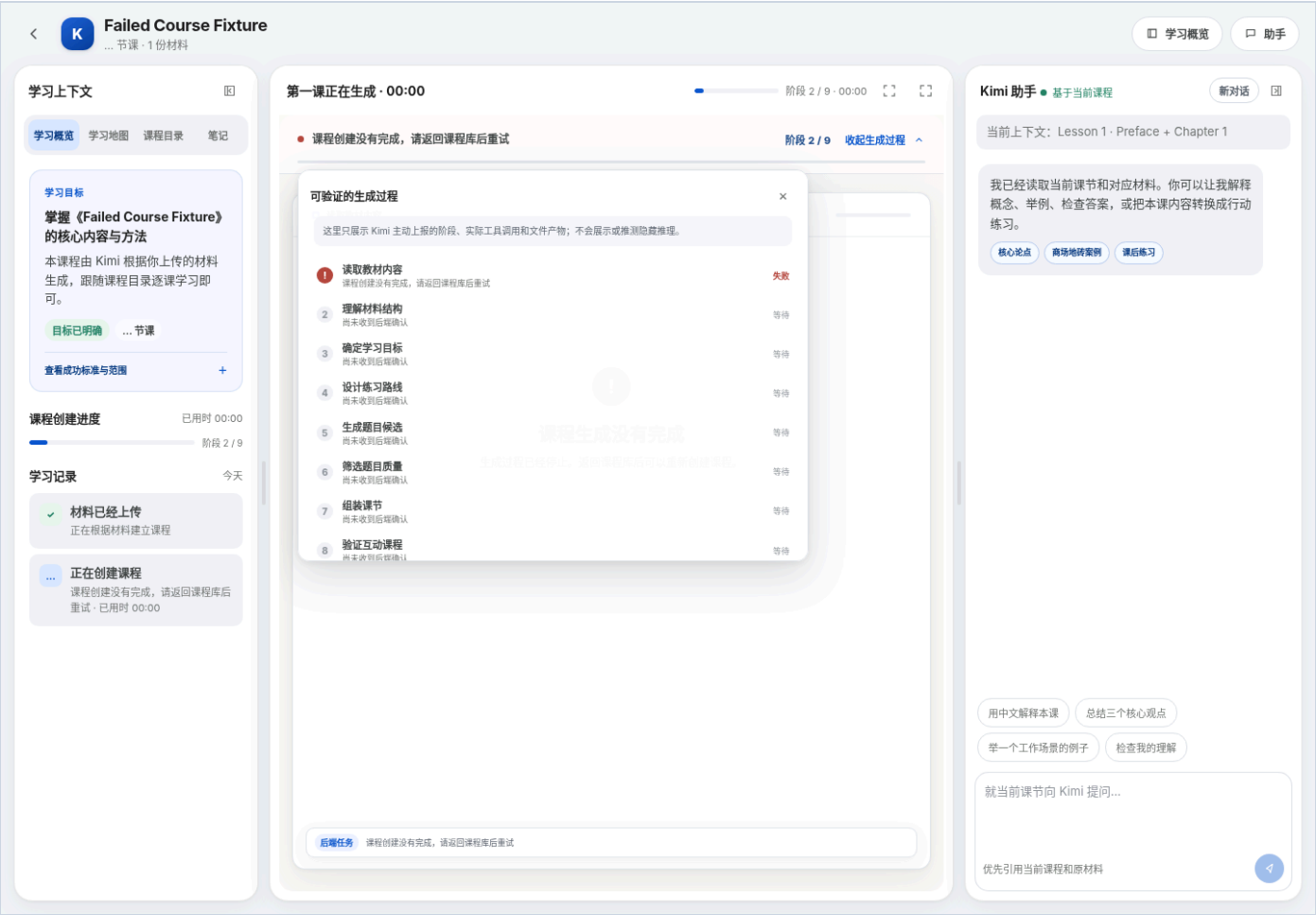
SS-002 - /course/failedcourse

PASS

路由

视口

时间



触发：

预期：

实际：

解释：“查看生成过程”可展开历史；Escape 收起后焦点返回触发器。

问题：

无

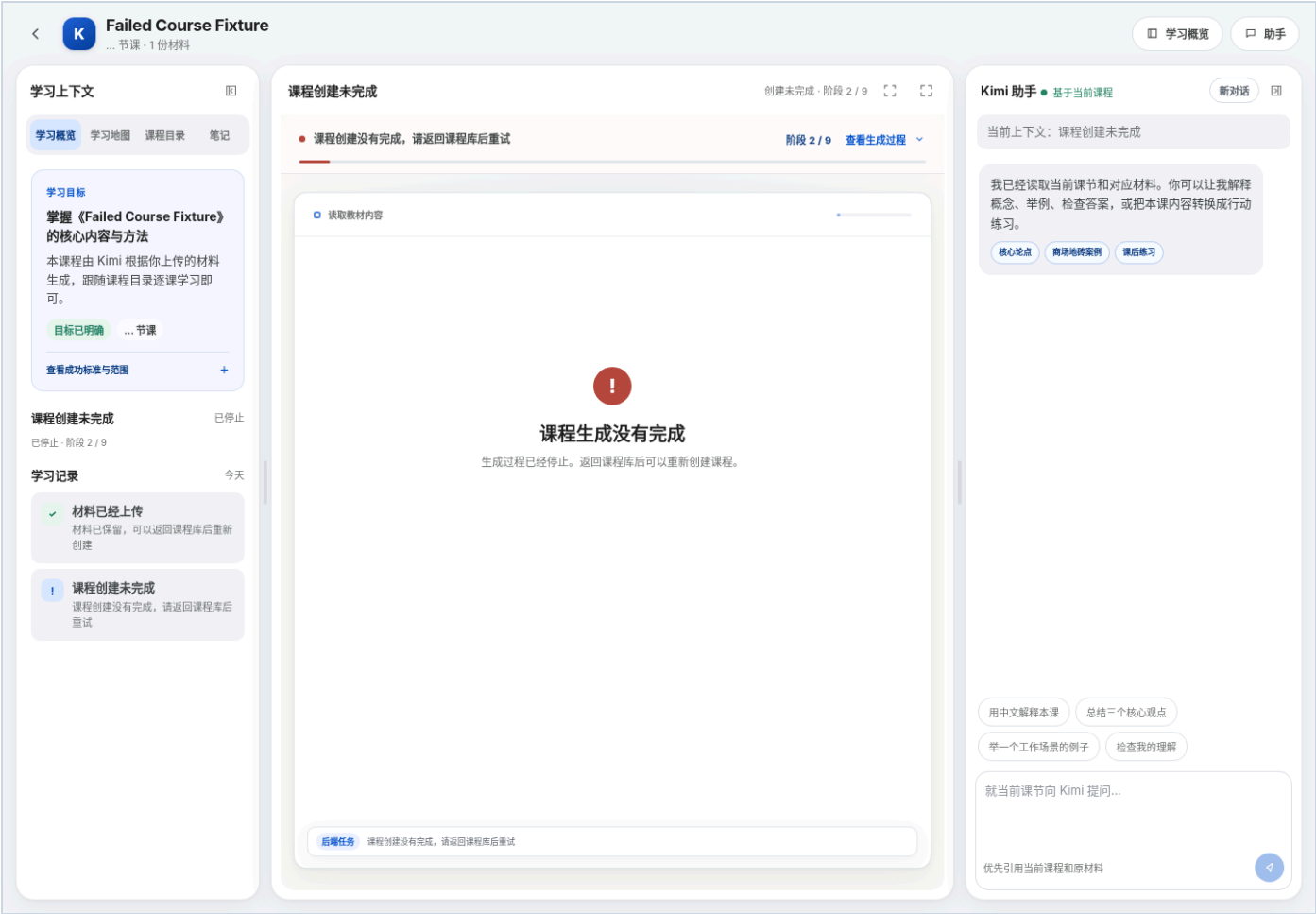
SS-003 - /course/failedcourse (repair copy)

PASS

路由

视口

时间



触发：

预期：

实际：

解释：所有可见状态面统一为失败/已停止；中央预览保留唯一 workflow-primary 进度条并显示后端最后进度 5%。

问题：

无

跨区域状态/进度审计

PASS

五个状态面生命周期一致；只有中央预览承担主进度，外围均为终态文本摘要。

区域	文案	实体	进度角色	标签	状态源	阶段	生命周期	%	置信度	关系证据
top-header	课程创建 未完成	course: failedc ourse	history_o nly			extractin g	failed		CONFIRM ED	
workspac e-header	创建未完 成 · 阶段 2 / 9	course: failedc ourse	history_o nly			extractin g	failed		CONFIRM ED	

区域	文案	实体	进度角色	标签	状态源	阶段	生命周期	%	置信度	关系证据
left-sidebar	课程创建 未完成 · 已停止 · 阶段 2 / 9	course: failedc ourse	history_o nly			extractin g	failed		CONFIRM ED	
main-preview	课程生成 没有完成	course: failedc ourse	workflow _primary	课程生成 进度	KimiGene rationPre view.upd ateProgre ss	extractin g	failed	5.0	CONFIRM ED	
left-activity	课程创建 未完成	course: failedc ourse	history_o nly			extractin g	failed		CONFIRM ED	

核对规则：

- same-entity lifecycle exclusivity
- active phase exclusivity
- progress agreement
- visual-state agreement
- action availability agreement

冲突：

无

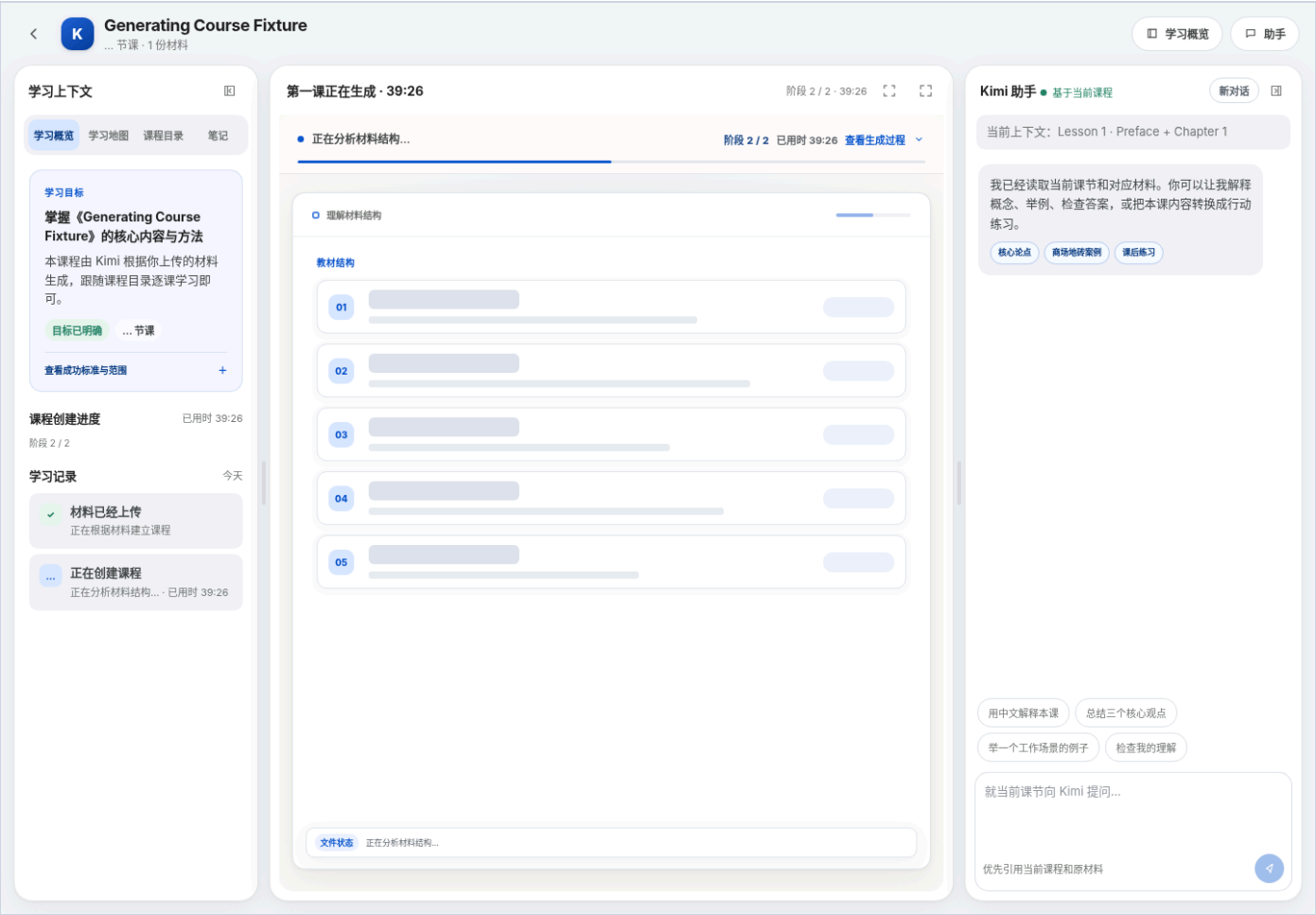
SS-004 - /course/generatingcourse (repair copy)

PASS

路由

视口

时间



触发：

预期：

实际：

解释：运行态仍保留生成标题和唯一中央主进度；页头/侧栏视觉进度条被降级为文本摘要。

问题：

无

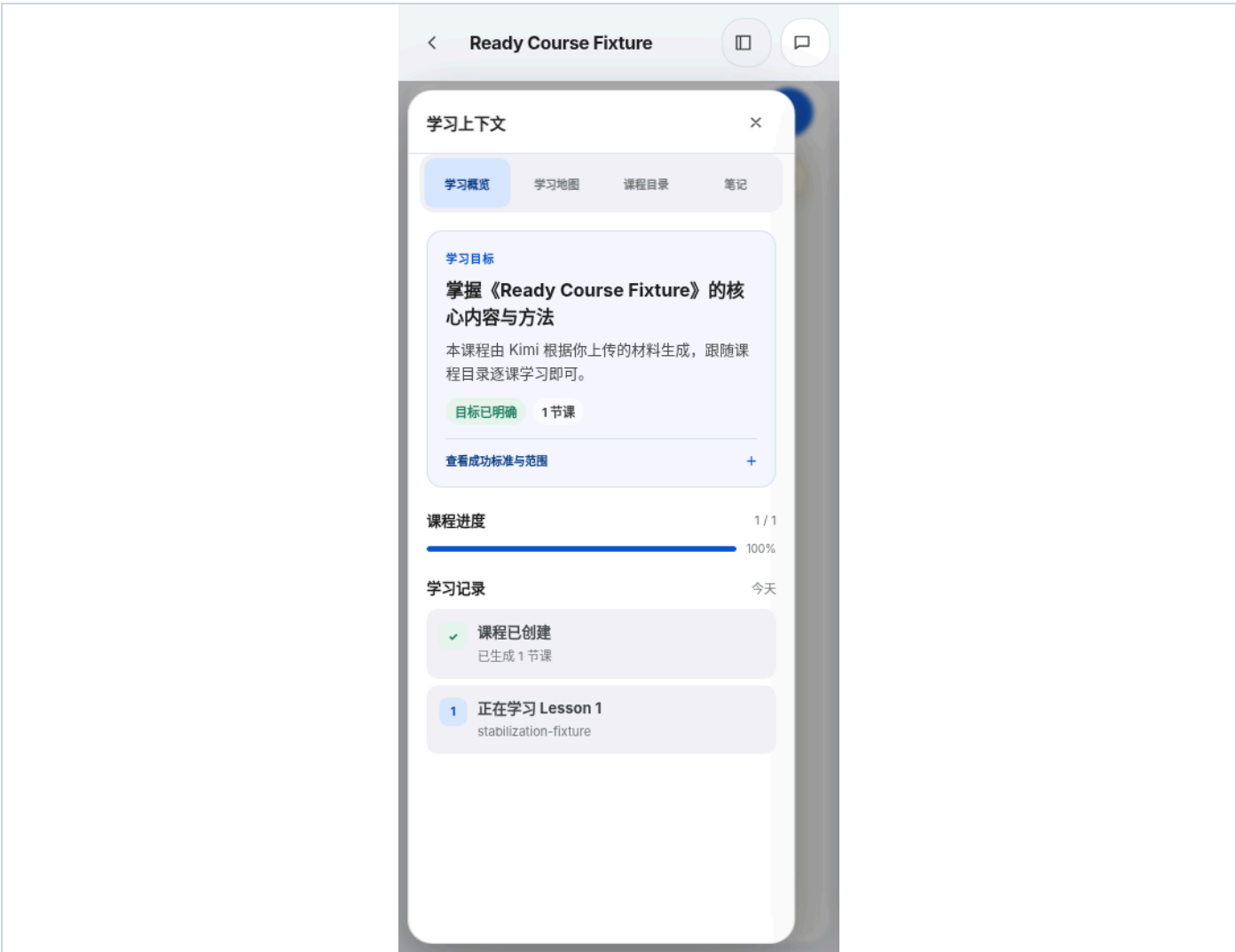
SS-005 - /course/readycourse 390x844

PASS

路由

视口

时间



触发：

预期：

实际：

解释：左抽屉、遮罩、44x44 下一课触控目标和无横向溢出已验证。

问题：

无

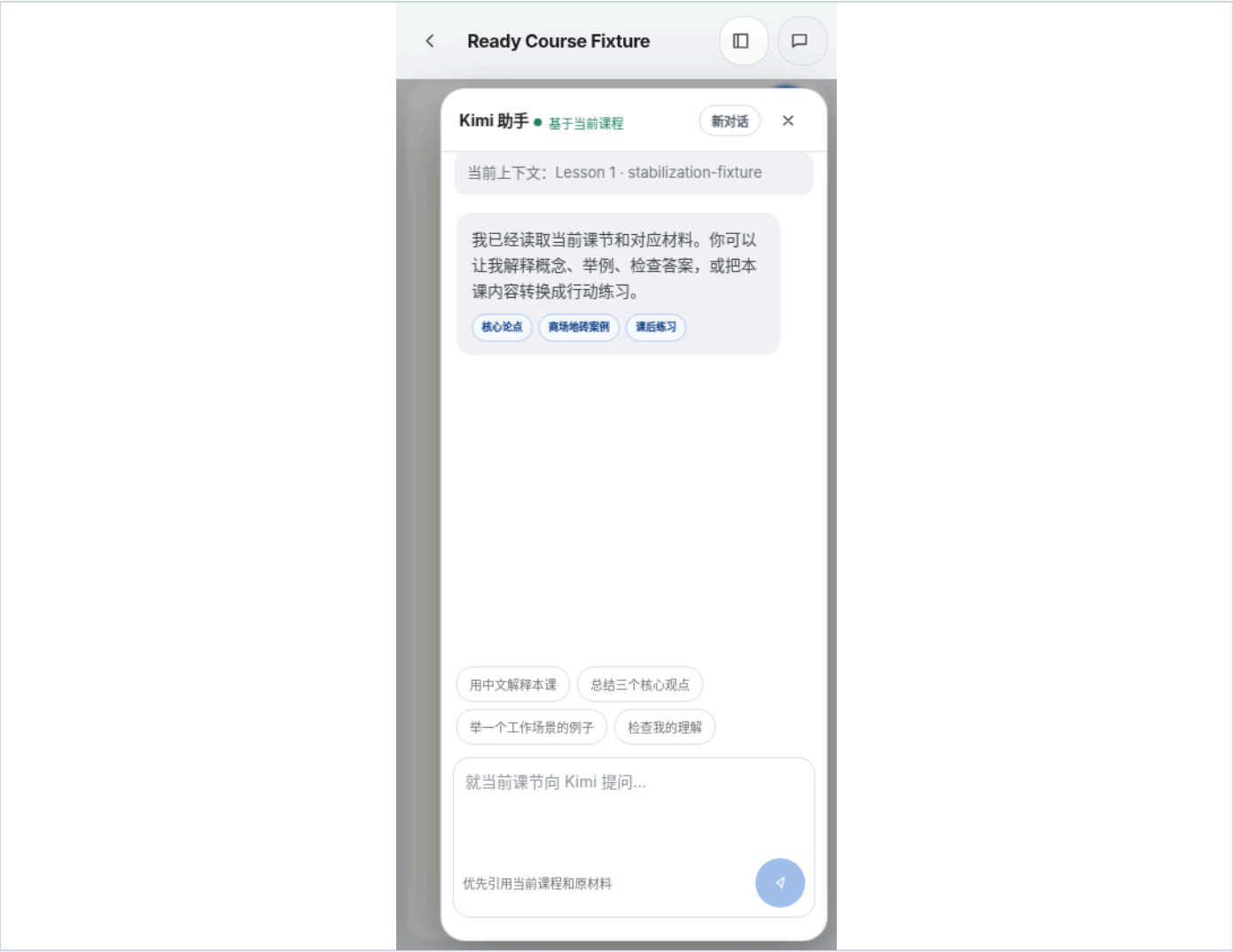
SS-006 - /course/readycourse 390x844

PASS

路由

视口

时间



触发：

预期：

实际：

解释：助手抽屉与学习概览互斥；Escape 关闭并把焦点还给顶部助手按钮。

问题：

无

9. 浏览器/网络错误

类型	页面	消息
Environment policy	browser startup	Chromium managed URLBlocklist=["*"] initially blocked all HTTP/HTTPS navigation; child-only policy isolation wrapper restored navigation without modifying system policy.

类型	页面	消息
External dependency	/course/readycourse Tutor	POST /api/courses/readycourse/chat -> 500 because kimi executable is absent.
External dependency	/course/failedcourse Tutor	POST /api/courses/failedcourse/chat -> 500 because kimi executable is absent.
Vendor asset	source viewer	GET /vendor/pdfs/build/pdf.mjs -> 404 in compatibility runtime because npm installation was unavailable.

10. 阻塞与未测试

路径	原因
App / Account header controls	No documented or implemented expected outcome; marked BLOCKED rather than inventing a verdict.
Tutor quick prompts and non-empty Send	Requires real Kimi CLI; local API produced explicit 500/ENOENT.
Generate next lesson success	Requires real Kimi CLI; failure recovery was observable, success/new lesson was not.
PDF/EPUB source success rendering	Vendor packages could not be installed due registry/network failures.

11. 改进路线图

立即行动

优先级	缺陷	行动	负责领域	工作量	依赖	验收	验证命令
P0	F-001	Apply F-001.patch and keep failed state atomic across all surfaces	generation state owner / frontend	S	none	- failed first frame has no running/busy surfaces - one primary progress bar - API progress reflected	python tests/e2e_py/generation_failure_chrome_test.py --base-url http://127.0.0.1:3108 --browser <chromium>
P0		Run Tutor/next-lesson integration gates with a sanctioned Kimi test binary before release	AI runtime integration	M	Kimi CLI fixture/test credential	- quick prompts and Send return Markdown answer - next lesson adds exactly one lesson - error path remains retryable	npm run test:e2e -- --grep "Tutor 下一课"

短期改进

优先级	缺陷	行动	负责领域	工作量	依赖	验收	验证命令
P1	F-001	Introduce one GenerationProgressView Model selector consumed by every renderer	state architecture	M	F-001 merged	- no renderer writes generation DOM independently - completion/failure transition is atomic	npm run test && npm run test:e2e -- --grep generation
P1		Define behavior contracts for App and Account header controls or remove them	product + frontend	S	product decision	each control has destination/state/permission/error behavior and E2E assertion	npm run test:e2e -- --grep header

长期改进

优先级	缺陷	行动	负责领域	工作量	依赖	验收	验证命令
P2	F-001	Add screenshot/state-snapshot invariant tests for every terminal lifecycle	quality engineering	M	canonical state selector	- ready/failed/interrupted screenshots reconcile header/sidebar/main/actions - duplicate primary progress automatically fails CI	<code>npm run test:e2e -- --grep "terminal state"</code>
P2		Vendor/cache browser and document-reader dependencies for hermetic CI	developer experience / CI	M	artifact registry	- npm install is not internet-dependent - PDF/EPUB smoke tests run offline	<code>npm ci --offline && npm run test:e2e -- --grep source</code>

12. 最终结论

本地 E2E 完成	True
发布决策	NO_GO
原因	原 ZIP 有 S2 失败终态冲突；此外 14 个外部依赖/未定义控件为 BLOCKED。修复候选已验证，但尚未应用回用户原始包。
最高严重度	S2 Major
第一个 Patch/阻塞	evidence/patches/F-001.patch
下一步	先应用 F-001.patch，再在有 Kimi CLI 与完整 vendor 依赖的环境重跑 Tutor、下一课和 Source Viewer 集成门禁。

残余风险

- 本报告的 AI 成功链没有真实模型响应证据。
- 兼容层仅用于隔离测试，不能替代正式 npm 依赖运行。
- ZIP 不含 Git commit，来源只能以 SHA-256 固定。
- 原始 ZIP 未被修改；修复只存在于 repair-worktree 和 patch 文件。